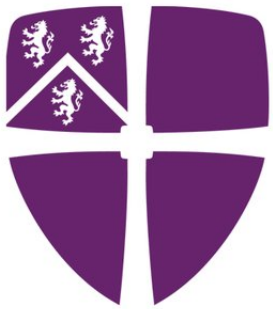
Twistor theory: a geometric approach to integrability

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Declaration of Authorship

This dissertation is a result of my own work through the review of the appropriate literature. I take no credit for the ideas presented herein and all sources I have consulted are referenced throughout.

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Introduction

Many theories in modern physics have some level of underlying geometry to them, the most famous example being Einstein's differential geometric view of gravity- general relativity. But even before then geometry had a major impact on physics with the likes of classical mechanics and its use of symplectic geometry. It is therefore not too surprising that there would be further attempts in using geometry to solve problems in theoretical physics. Indeed, in 1967 Roger Pensrose introduced a new geometric theory called twistor theory [1] in an attempt to develop a theory of quantum gravity. The theory had rich geometric insights and has through the years gave us advances in many areas of physics. Unfortunately, twistor theory has not yet realised Penrose's dream for a path to quantum gravity but it has instead helped the study of integrable systems, instantons and, most recently, scattering amplitudes tremendously.

Penrose's motivations were mainly fuelled by his opinion that the description of spacetime at the time was inadequate [2]. Indeed, the accepted view of space-time was that quantum mechanics can affect its structure at only small, quantum gravitational, distances, but this did not lead to any reconcilliation of general relativity and quantum mechanics. As Penrose states:

> *"While it might be that only at 10^{-33} cm is it necessary to invoke a description of space-time radically removed from that of a manifold, my view was (and still is) that even at the much larger levels of elementary particles, or perhaps atoms, where quantum behaviour holds sway, the standard space-time descriptions have ceased to be the most physically appropriate ones."*

Penrose then pulled from many areas of mathematics to form his new theory. In particular, the Klein correspondence and the Radon transform were seen as main motivators in this sense, but areas far removed from geometry, such as spin-network theory, influenced him as well.

Following the early work of Penrose, the theory developed rapidly to encompass several areas of mathematics. This progression lead to several results in the field of Riemannian geometry and integrable systems. We will mostly focus on the latter, where twistor theory has had the most notable impact.

This report can be seen as being split into three distinct parts. The first of which introduces the mathematical background required for us to flesh out twistor theory makes up the first two chapters while the next two walk us through twistor theory's application to our area of interest-integrability in mathematical physics. The final chapter, excluding the conclusion, will see us briefly discuss the generalisation of twistor theory to curved spacetimes.

We start chapter 1 by introducing the theory of bundles which will find use throughout the report. Moving on, we look at the notions of duality and spinors before discussing sheaf theory and its associated cohomology theory. Some of these notions will then be used in chapter 2 to form two different interpretations of the twistor space before moving on to some of the major results of the theory.

The second part, consisting of chapter 3 and 4, will discuss the zero-rest mass equations and the anti-self dual Yang-Mills equations respectively. Both of these will showcase the techniques from chapter 1 in conjunction with the twistor notions from chapter 2 to prove integrability theorems for both sets of equations. The end of chapter 4 then notes the application of our discussion to the study of instantons and symmetry reductions.

Chapter 5 will discuss the generalisations of twistors to curved spacetimes and, perhaps more importantly, showcase the difficulties in this process. We will detail two possible general-

isations; ambitwistors and anti-self dual spacetimes. For the latter of which we will motivate a powerful, yet limited, theorem which will allow us generate such spacetimes from appropriate spacetimes.

1 Geometric and topological concepts

In this section we will present many of the geometric and topological ideas that arise in twistor theory. We will start by defining the notion of a bundle before moving onto the subject of sheaf cohomology, which we will use extensively in chapter 3. We aim to give an intuitive understanding of the concepts rather than a mathematically rigourous one, so we will refer the reader to appropriate texts throughout if more context is needed. In particular, much of the bundle theory is motivated by section 7 of Nash & Sen [3].

1.1 Fibre bundles and special cases

When studying the properties of topological spaces it is often useful to try and break them up into products of more elementary spaces. Of course, this cannot always be done globally so the best we can hope for in most cases is some form of local decomposition. A *fibre bundle* formalises this notion. Due to thus, bundles have found great use in the likes of topology, geometry and even mathematical physics where they either simplify certain situations or allow us to formulate useful algebraic structures over a so-called *base space*. This leads us to the following definition.

Definition 1.1. A fibre bundle is a topological space E, called the *total space*, together with two other topoligcal spaces B, the *base space*, and F, the *fibre*, equipped with a projection $\pi : E \to B$. Furthermore, we require that for each $x \in E$ there is an open neighbourhood $U \subset B$ of $\pi(x)$ such that there are homeomorphisms $\varphi : \pi^{-1}(U) \to U \times F$ that obey the following commutative diagram

$$\begin{array}{ccc} \pi^{-1}(U) & \xrightarrow{\varphi} & U \times F \\ {\scriptstyle \pi}\downarrow & \swarrow_{\mathrm{proj}_U} & \\ U & & \end{array}$$

We call (U, φ) a *local trivialisation* and typically denote a fibre bundle by $\pi : E \to B$.

There are two objects not included in the definition of a fibre bundle that we will find very useful. The first of which is familiar from the study of manifolds- *transition functions*. Say we have two open sets U_i and U_j of B such that $U_{ij} = U_i \cap U_j \neq \emptyset$. Then the transfer function T_{ij} is defined by $T_{ij} = \varphi_j \circ \varphi_i^{-1}$ where (U_i, φ_i) is a local trivialisation of U_i. The transition function t_{ij} corresponding to this is then

$$T_{ij}(x, f) = (x, t_{ij} f).$$

Intuitively, a transition function encodes the twisting of the bundle and a bundle with constant transition functions is said to be flat.

The second object we require is that of a *section*. A section of a fibre bundle $\pi : E \to B$ is a function $\gamma : B \to E$ that obeys $\pi \circ \gamma = \mathrm{id}_B$. A section smoothly assigns each point in the base space a point in the total space as can be seen in figure 1. Global sections don't always exist so we often have to resort to using sections defined only on an open subset of B and then gluing these together to form a section over the whole space. Using our, possibly local, sections

we can therefore pullback objects defined on the total space to the base space and generalise our base space objects to the total space. We denote the space of sections of a fibre bundle $\pi : E \to B$ by $\Gamma(B, E)$ or simply $\Gamma(E)$.

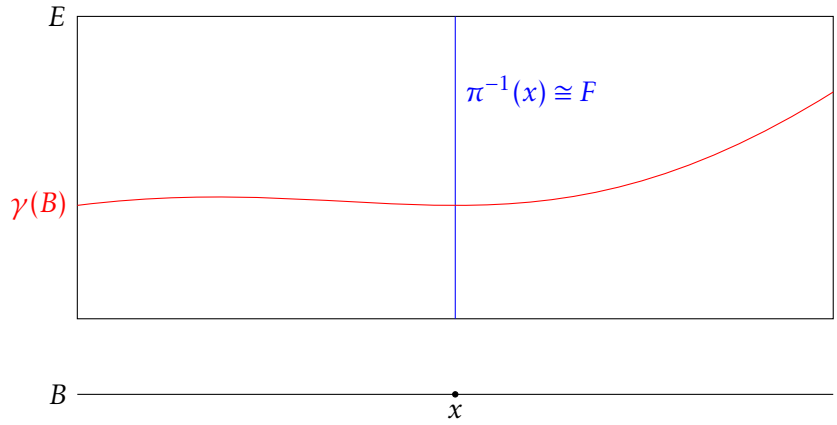


Figure 1: A demonstration of how a section γ smoothly assigns each point in the base space B a point in the total space E. Notice that it intersects the fibre over a point $x \in B$ once exactly.

The definition of a fibre bundle is intentionally very general. Indeed, the most important bundles that we will encounter belong to certain special cases of the general definition. The most prominent of these are *vector bundles* and *principal bundles* which we will now define.

Definition 1.2. A *vector bundle* is a fibre bundle $\pi : V \to B$ such that for any point $x \in B$, the fibre $\pi^{-1}(x)$ is a vector space. We say that the vector bundle is of rank-n if each fibre is n-dimensional. A rank-1 vector bundle is called a *line bundle*.

Definition 1.3. A *principal G-bundle* is a fibre bundle $\pi : P \to B$ together with a Lie group G that acts both freely and transitively on fibres while also preserving them. That is, for any $y \in \pi^{-1}(x)$ we have $y \cdot g \in \pi^{-1}(x)$ for all $g \in G$. We call G the *structure group* or *gauge group*.

There are two things we must note about principal bundles. Firstly, we have made the convention that principal bundles are acted upon from the right by their structure group but we could have equally chosen the group to have a left action. Secondly, the definition of a principal bundle itself implies that the fibres are homeomorphic to the structure group.

We now discuss three examples which highlight the notions we've introduced above. In particular, we will introduce the tautological line bundle and its extensions for which we will use later in the study of sheaf cohomology.

Example 1.1. For a manifold $\mathcal{M}$, the *tangent space* $\mathrm{T}_x\mathcal{M}$ at a point $x \in \mathcal{M}$ is simply the collection of tangent vectors at the point x. We can construct the *tangent bundle* $\mathrm{T}\mathcal{M}$ as follows:

$$\mathrm{T}\mathcal{M} = \bigsqcup_{x \in \mathcal{M}} \mathrm{T}_x\mathcal{M} = \bigcup_{x \in \mathcal{M}} \{x\} \times \mathrm{T}_x\mathcal{M}.$$

It is clear that the tangent bundle is not only a fibre bundle, but in actuality a vector bundle. Sections of the tangent bundle are vector fields and the dual bundle is the *cotangent bundle* $\mathrm{T}^*\mathcal{M}$, whose sections are 1-forms.

Example 1.2. Let $\mathcal{M}$ be an n-dimensional manifold. Now let $\mathrm{F}_x\mathcal{M}$ be the set of ordered bases of $\mathrm{T}_x\mathcal{M}$. The *frame bundle* $\mathrm{F}\mathcal{M}$ is defined to be

$$\mathrm{F}\mathcal{M} = \bigsqcup_{x\in\mathcal{M}} \mathrm{F}_x\mathcal{M}.$$

Now we observe that if $\boldsymbol{e} = (e_1, \ldots, e_n)$ is a basis of $\mathrm{T}_x\mathcal{M}$, then so is $\boldsymbol{e}A$ for $A \in \mathrm{GL}(n)$. We therefore see that $\mathrm{GL}(n)$ preserves the fibres of $\mathrm{F}\mathcal{M}$. Indeed, the frame bundle of a manifold $\mathcal{M}$ is a $\mathrm{GL}(\dim\mathcal{M})$-principal bundle.

Example 1.3. We define the n-dimensional *complex projective space* $\mathbb{CP}^n$ as

$$\mathbb{CP}^n = \{\text{one-dimensional subspaces of } \mathbb{C}^{n+1}\}.$$

It can be easily shown that $\mathbb{CP}^n$ has the structure of a manifold. The *tautological line bundle* L^{-1} fibres over $\mathbb{CP}$ by associating to each point the one dimensional line it represents in $\mathbb{C}^2$. Using L^{-1} we can construct other line bundles over $\mathbb{CP}$ by defining the following bundles

$$L^{-k} = (L^{-1})^{\otimes k} \qquad L^k = (L^{-k})^* \quad \text{and} \quad L^0 = L^{-1} \otimes L^1.$$

We note that the sections of L^k, typically denoted by $\mathcal{O}(k)$, can be regarded as a homogeneous function of degree k on $\mathbb{C}^2$. The derivation of this fact is somewhat tedious, so we instead refer readers to appendix B of Dunajski [4] for a description of this.

Since a principal bundle carries a structure group it is natural to ask what type of structure we can form given a representation of the group. So let $\pi : P \to B$ be a principal G-bundle, F some space that G acts upon and $\rho : G \to \mathrm{Aut}(F)$ be the corresponding representation[1]. This information allows us to define a fibre bundle over the total space

$$P \times_G F := (P \times F)/G$$

where dividing by the structure group gives us the identification $(p, f) \sim (p \cdot g, \rho(g^{-1})f)$. This induces a natrual projection $\hat{\pi} : P \times_G F \to B$ by taking $\hat{\pi}(p, f) = \pi(p)$. We have therefore constructed a fibre bundle that is said to be *associated to* P *through* ρ. This construction allows us to tackle problems involving principal bundles by instead using the associated fibre bundle, which often turns out to have a vector bundle structure.

Example 1.4. Let $\pi : P \to B$ be a principal G-bundle and $\mathfrak{g}$ the corresponding Lie algebra to the structure group. Suppose that G is a matrix group. Then the adjoint action on the Lie algebra is given by

$$\mathrm{ad}_g : \mathfrak{g} \to \mathfrak{g}, \quad \mathrm{ad}_g Z = gZg^{-1},$$

for any $g \in G$. The *adjoint bundle* is then defined to be $\mathrm{ad}P = P \times_G \mathfrak{g}$ and is associated to P through the adjoint representation.

[1] To clarify, $\mathrm{Aut}(F)$ is the group of automorphisms on the space F.

Having introduced all of our bundle technology, we now pose an important question: when is a bundle trivial? By this we mean, when is a bundle equivalent to the *trivial bundle* defined by $E = B \times F$? If the bundle is trivial then the total space is globally a product space and is therefore much simpler to work with. There has been much work put into answering this question, with several results being obtained for principal bundles and vector bundles, the most important of which we now state, from appendix B of Dunajski [4], for later use.

Proposition 1.1. A rank-k vector bundle is trivial if and only if there exist *splitting matrices* $h_a : U_a \to \mathrm{GL}(k)$ that obey

$$f_{ab} = h_b h_a^{-1}$$

where f_{ab} are the transition functions of the vector bundle.

Now that we've completed our introduction to bundles (mostly), the reader may wonder how these structures appear in mathematical physics. While the objects above found their start in the study of geometry and topology, it was not long before the attractive qualities of bundles found their way into several parts of physics. For example, we will see in chapter 4 that the natural formulation of the Yang-Mills equations involve an argument based on exterior forms on principal bundles where gauge transformations are viewed as G-equivariant automorphisms compatible with the base space projection. It should also be noted that these notions also manifest themselves very naturally in general relativity and, while we will not go into detail here, we refer readers to [5] for more details.

1.2 The Hodge star operator

An important operator defined on manifolds and bundles is the *Hodge star operator*. It is important as it allows us to give exterior forms a notion of duality and, as explained in [6], it is one of three isomorphisms which make twistor theory particularly useful in 4-dimensions.

Let $(\mathcal{M}, g)$ be a psuedo-Riemannian manifold[2] with a volume form ω[3] and let $\Lambda^p(\mathcal{M})$ denote the space of p-forms on the manifold. The metric induces a natural bilinear form on the space of p-forms

$$\langle \cdot, \cdot \rangle : \Lambda^p(\mathcal{M}) \otimes \Lambda^p(\mathcal{M}) \to \Lambda^0(\mathcal{M}).$$

Provided that $\mathcal{M}$ is compact, we can define a global inner product given by

$$(\vartheta_i, \vartheta_j) = \int_{\mathcal{M}} \langle \vartheta_i, \vartheta_j \rangle \, \omega.$$

This is called the *Hodge inner product*. If $\mathcal{M}$ is n-dimensional, the *Hodge star operator* is defined by

$$\star : \Lambda^p(\mathcal{M}) \to \Lambda^{n-p}(\mathcal{M}), \quad \vartheta_i \wedge \star \vartheta_j = \langle \vartheta_i, \vartheta_j \rangle \, \omega.$$

[2] That is, a smooth manifold $\mathcal{M}$ equipped with a smooth, non-degenerate, symmetric metric g.

[3] From here on out we take $\omega = \sqrt{|\det g|}\, e^1 \wedge \ldots \wedge e^n$ where $\{e^1, \ldots, e^n\}$ is a basis of the cotangent space of at some point in $\mathcal{M}$ whenever we specify a coordinate basis.

Note that we call $\star\vartheta$ the *Hodge dual* of ϑ.

While we have defined the Hodge star in a coordinate free manner, we can write down a component formula for it. Let $\{e_1,...,e_n\}$ be a local basis for $T\mathcal{M}$ and denote the dual basis by $\{e^1,...,e^n\}$. Then for a p-form $\vartheta = \frac{1}{p!}\vartheta_{i_1...i_p}e^{i_1}\wedge...\wedge e^{i_p}$ we have

$$\star\vartheta = \frac{1}{p!(n-p)!}\epsilon^{j_1...j_p}{}_{i_{p+1}...i_n}\sqrt{|\det g|}\ \vartheta_{j_1...j_p}e^{i_{p+1}}\wedge...\wedge e^{i_n},$$

which gives us a more practical means of computing $\star\vartheta$. Using this (and the definition) we have the following useful consequences:

1. For a p-form ϑ on an n-dimensional manifold we have
$$\star(\star\vartheta) = \mathrm{sig}(g)(-1)^{p(n+1)}\vartheta$$
2. Given another p-form ε we must have
$$(\vartheta,\varepsilon) = (\star\vartheta,\star\varepsilon).$$
3. Finally, the Hodge dual of the unit 0-form is the volume form. That is, $\star 1 = \omega$.

An important case of Hodge duality occurs when we are working on a manifold with even dimension. For this discussion we will take the dimension of the discussed manifolds to be 4 as this is the most physically relevant, but the discussion is easily generalisable.

On a 4-dimensional manifold $\mathcal{M}$ we notice that the action of the Hodge star on 2-forms is particularly interesting. Namely, the Hodge star now acts as a map from $\Lambda^2(\mathcal{M})$ to itself. Furthermore, it acts as an involution (up to a minus sign) as applying it twice simply gives us the identity map up to a minus sign. This induces a natural decomposition into the eigenspaces of the Hodge star:

$$\Lambda^2(\mathcal{M}) = \Lambda^2_+(\mathcal{M})\oplus\Lambda^2_-(\mathcal{M})$$

where

$$\Lambda^2_\pm(\mathcal{M}) = \{\text{2-forms } F | \star F = \pm\sqrt{\mathrm{sig}(g)}\ F\}.$$

Forms that obey $\star F^+ = \sqrt{\mathrm{sig}(g)}\ F^+$ are called ***self dual*** while forms that obey $\star F^- = -\sqrt{\mathrm{sig}(g)}\ F^-$ are called ***anti-self dual.*** Take, for example, the case of a Lorentzian metric where $\mathrm{sig}(g) = -1$. Then the above decomposition can be explicity written as

$$F = \underbrace{\tfrac{1}{2}(F + i\star F)}_{F^-} + \underbrace{\tfrac{1}{2}(F - i\star F)}_{F^+}.$$

By a simple calculation we see that the first term is anti-self dual while the second is self dual.

Finally we note that when dealing with the inner products we've introduced, we regularly encounter situations where the exterior derivative is acting on one of the arguments. It is often advantageous, or necessary, to move this operator to the other argument via its adjoint with respect to the inner product. The Hodge star operator gives us a way of doing so via the following definition and proposition.

Definition 1.4. Let $\mathcal{M}$ be an n-dimensional manifold equipped with a metric g. The *codifferential* δ operating on a p-form is defined to be

$$\delta = \mathrm{sig}(g)(-1)^{pn+1} \star \mathrm{d} \star .$$

Proposition 1.2. The codifferential defined above is the formal adjoint of the exterior derivative d. That is,

$$(\mathrm{d}\vartheta, \varepsilon) = (\vartheta, \delta\varepsilon)$$

for $\vartheta \in \Lambda^p(\mathcal{M})$ and $\varepsilon \in \Lambda^{p+1}(\mathcal{M})$.

Proof. The proof follows from simple calculation. We first note that

$$\mathrm{d}(\vartheta \wedge \star\varepsilon) = \mathrm{d}\vartheta \wedge \star\varepsilon + (-1)^p \vartheta \wedge \mathrm{d}(\star\varepsilon).$$

Now we insert a factor of $1 = \mathrm{sig}(g)(-1)^{p(n+1)} \star\star$ into the second term to get

$$\mathrm{d}(\vartheta \wedge \star\varepsilon) = \mathrm{d}\vartheta \wedge \star\varepsilon + \mathrm{sig}(g)(-1)^p(-1)^{p(n+1)} \vartheta \wedge \star(\star \mathrm{d}(\star\varepsilon)).$$

This allows us to insert the codifferential in the final term to get

$$\mathrm{d}(\vartheta \wedge \star\varepsilon) = \mathrm{d}\vartheta \wedge \star\varepsilon - \vartheta \wedge \star\delta\varepsilon.$$

The result follows by integrating each side and using Stokes' theorem to show the left-hand side vanishes. □

This proposition will be used later in chapter 4 when dealing with gauge fields but the codifferential has a rich and varied use in Riemannian geometry and *Hodge theory* where it is used to construct partial differential equations whose solutions tell us fundamental properties of the underlying manifold. A brief introduction to Hodge theory and the use of the codifferential can be found in [7].

1.3 Spinor bundles

Spinors, originally formulated geometrically by Cartan in 1913, have found unprecedented use in many areas of modern physics and twistor theory is no exception. Although they are familiar from field theory, a more complete understanding can be given through bundles. Here we outline our conventions and discuss some of the bundle theory of spinors. We will, however, not delve too far into the theory of spin bundles and instead refer readers to [8], for a rigourous treatment, and [9], for a brief yet detailed overview.

In Minkowski space $\mathbb{M}$ we can construct an isomorphism between vectors and 2×2 Hermitian matrices. Explicitly the map $\varphi : \mathbb{M} \to H_{2\times 2}$ can be given as

$$\varphi(x^a) = \frac{1}{\sqrt{2}} \begin{pmatrix} x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & x^0 - x^3 \end{pmatrix}. \tag{1.1}$$

If we instead consider the domain of φ to be *complexified Minkowski space* $\mathbb{CM} = \mathbb{M} \otimes \mathbb{C}$ then the image of φ is the space of 2×2 complex matrices $\mathbb{C}^{2\times 2} \cong \mathbb{C}^4$. By choosing a basis and

origin for $\mathbb{M}$ we can identify it with its tangent space at the origin and therefore, by symmetry, identify this with the tangent space at any point $p \in \mathbb{CM}$. We therefore have the following isomorphisms

$$\mathrm{T}_p\mathbb{CM} \cong \mathbb{C}^4 \cong S \otimes S^*,$$

where $S = \mathbb{C}^2$ and S^* is the space dual to S. The space S the is called the ***spinor space*** of $\mathbb{CM}$ at p. We can now form a bundle in the usual way by taking the disjoint union of this relation over all points in $\mathbb{CM}$ to get

$$\mathrm{T}\mathbb{CM} \cong \mathbb{S} \otimes \mathbb{S}^*,$$

where $\mathbb{S}$ is called a ***spinor bundle***.

The spinor bundle is in fact an $\mathrm{SL}(2,\mathbb{C})$-bundle and, as such, there must be a non-degenerate skew-symmetric form $\epsilon \in \Gamma(\mathbb{CM}, \bigwedge^2 \mathbb{S})$ which we call a ***symplectic form***[4]. This symplectic form helps us describe the sections of the spinor bundle, aptly called spinor fields, locally and endow them with a consistent index notation. The index notation can be summarised as follows:

$$\begin{array}{ccc} \pi^A \in \Gamma(\mathbb{CM}, \mathbb{S}) & \xrightarrow{\times \epsilon_{BA}} & \pi_B \in \Gamma(\mathbb{CM}, \mathbb{S}^*) \\ \text{conjugation}\downarrow & & \downarrow \\ \bar{\pi}^{A'} \in \Gamma(\mathbb{CM}, \bar{\mathbb{S}}) & \xrightarrow{\times \epsilon_{B'A'}} & \bar{\pi}_{B'} \in \Gamma(\mathbb{CM}, \bar{\mathbb{S}}^*) \end{array}$$

where we see that the symplectic form acts as a map into the duals of the spinor spaces. In this notation, equation 1.1 can simply be stated as $x^{AA'} = x^a \sigma_a{}^{AA'}$ where σ_a is the vector of Pauli matrices which naturally carries mixed indices. We will often use the convention that we neglect the σ_a and simply write $x^a = x^{AA'}$. Furthermore, we take the symplectic form ϵ to be given by

$$\epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

while doing actual computation.

The index notation extends very simply to higher valence spinors where we formally define an n-spinor as

$$\pi^{A_1 A_2 \ldots A_n} \in \Gamma(\mathbb{CM}, \underbrace{\mathbb{S} \otimes \ldots \otimes \mathbb{S}}_{n \text{ many copies}})$$

and analogously for mixed and lowered indices. When the valence of a spinor is two or greater, we have certain simplifications due to the two-dimensionality of the spinor space. In particular any two-spinor must satisfy

$$2\pi^{[AB]} := \pi^{AB} - \pi^{BA} = \epsilon^{AB} \pi_C{}^C$$

[4]The existence of this form on a general spinor bundle arises from the fact that an $\mathrm{SL}(2,\mathbb{C})$-bundle is a reduction of a $\mathrm{GL}(2,\mathbb{C})$-bundle if and only if there is a choice of volume form. For more information see section 2.3 of Ward & Wells [10].

due to the fact that there is only one antisymmetric two-dimensional matrix up to scaling. This tells us that anti-symmetric spinors are *pure trace* and any antisymmetric indices of an n-spinor can be removed through contracting the indices and multiplying by the symplectic form. We also note that the relative ordering of dashed and undashed indices is irrelevant.

The isomorphism introduced in equation 1.1 also extends naturally, giving an isomorphism between n-tensors and equally mixed index $2n$-spinors by applying the map to each index. A particularly useful example is that the metric η_{ab} in spinor notation is given by

$$\eta_{ab}\sigma^{a}{}_{AA'}\sigma^{b}{}_{BB'} = \epsilon_{AB}\epsilon_{A'B'}.$$

This notation also proves useful when discussing the Hodge duality of tensors. To illustrate this, consider a two-form R_{ab} with the equivalent spinor $R_{AA'BB'}$. Then we must have

$$R_{AA'BB'} = \tfrac{1}{2}(R_{AA'BB'} + (R_{AB'BA'} - R_{AB'BA'}) - R_{BB'AA'}) = \tfrac{1}{2}(R_{AB[A'B']} - R_{[BA]B'A'}).$$

As discussed above, we can extract the anti-symmetric indices by taking a trace. Doing so gives us

$$R_{AA'BB'} = \tfrac{1}{2}R_{ABC'}{}^{C'}\epsilon_{A'B'} + \tfrac{1}{2}R_{A'B'C}{}^{C}\epsilon_{AB} := \phi_{AB}\epsilon_{A'B'} + \phi_{A'B'}\epsilon_{AB}.$$

It is not hard to see that the two-spinors ϕ_{AB} and $\phi_{A'B'}$ are symmetric in their respective indices. For example, we have that

$$\phi_{[AB]} = \tfrac{1}{4}R_{CC'}{}^{CC'}\epsilon_{AB} = \tfrac{1}{4}\epsilon_{AB}R_{a}{}^{a} = 0,$$

by the anti-symmetry of R_{ab}. In this notation, the duality of the tensor is manifest. Indeed, it can be easily checked that $\phi_{A'B'}\epsilon_{AB}$ is the self dual part of R_{ab} while $\phi_{AB}\epsilon_{A'B'}$ is the anti-self dual part.

1.4 Sheaf theory

The final ingredient we need to construct the most major results in twistor theory is that of *sheaf cohomology*. Mathematically, sheaf theory tells us to which extent information on an open set can be learned from a restriction of that set, leading to some calling it a theory of locality. We, however, will use it to interpret twistor transformations in terms of cohomology groups, therefore giving rise to many of the major results in twistor theory. Our overall style and conventions for this section will mostly follow the text of Huggett and Todd [11].

Definition 1.5. Let M be a topological space. A *presheaf* $\mathcal{F}$ of abelian groups on M is an assignment of an open set $U \subset M$ to an abelian group $\mathcal{F}(U)$. Furthermore, we require that for any two open sets $U_j \subset U_i \subset M$, there are *restriction homomorphisms*

$$r^{U_i}_{U_j} \equiv r^{i}_{j} : \mathcal{F}(U_i) \to \mathcal{F}(U_j)$$

with the property that for open sets $U_k \subset U_j \subset U_i$

$$r^{i}_{j} \circ r^{j}_{k} = r^{i}_{k} \quad \text{and} \quad r^{i}_{i} = \mathrm{id}.$$

Technically speaking, we are not restricted to talking about presheaves of abelian groups. We could speak of presheaves of rings, modules over a ring and so on. The main point is that a presheaf is a collection of objects of the same type, paramaterised by open sets. Compare this to a bundle, which is a collection of objects of the same type parameterised by the points of a space.

For many purposes a presheaf is *too* general. We often require some additional structure and therefore must impose conditions upon it. Namely, we want to be able to determine elements of presheaves by their local behaviour and be able to glue local elements together to form global elements. By demanding this we get the following definition.

Definition 1.6. A *sheaf* is a presheaf $\mathcal{S}$ satisfying the follwing two properties:

1. If $\{U_a\}$ is a collection of open sets in M, where $U = \bigcup U_a$ and $U_{ij} = U_i \cap U_j$[5], $s_a \in \mathcal{S}(U_a)$ and

   $$r^a_{ab}(s_a) = r^b_{ab}(s_b)$$

   for all a and b, then there exists an $s \in \mathcal{S}(U)$ such that

   $$r^U_{U_a}(s) = s_a, \quad \text{for all } a.$$

2. If $\{U_a\}$ is a collection of open sets in M, where $U = \bigcup U_a$, and if s and $s' \in \mathcal{S}(U)$ satisfy

   $$r^U_{U_a}(s) = r^U_{U_a}(s')$$

   for all a, then $s = s'$.

Elements of a sheaf $\mathcal{S}(U)$ are called *sections*. As the name may suggest, the sections of a fibre bundle form a sheaf and the most important sheaves we consider arise in this fashion. Let us now list some important sheaves that we will need for later use.

Example 1.5. Let $\mathcal{M}$ be a complex manifold and $\pi : E \to \mathcal{M}$ be a holomorphic vector bundle that fibres over $\mathcal{M}$. We can then define the *structure sheaf* of $\mathcal{M}$ by

$$\mathcal{O} = \{\mathcal{O}(U) = \text{holomorphic functions on } U \subset \mathcal{M}\}.$$

Note that $\mathcal{O}$ is an example of a sheaf of rings. Using this we can define the following sheaves of modules over $\mathcal{O}$:

$$\begin{aligned} \Omega^p &= \{\Omega^p(U) = \text{holomorphic } p\text{-forms on } U\}, \\ \mathcal{O}(E) &= \{\mathcal{O}(U, E) = \text{holomorphic sections of } E \text{ over } U\}, \\ \Omega^p(E) &= \{\Omega^p(U, E) = E\text{-valued } p\text{-forms on } U\}. \end{aligned}$$

As mentioned earlier, sheaf theory can tell us about the locality of certain information. To this end, we wish to define an object which tells us about the localisation of information about

[5]We shall use this notation throughout the rest of the report.

a point. To do so, we let $\mathcal{S}$ be a presheaf over a topological space M and let $U(x)$ denote the set of open sets of M that contain x. We then define

$$\Sigma_x = \bigsqcup_{U(x)} \mathcal{S}(U).$$

This set unfortunately has a redundancy due to the fact that the restriction of elements from different sets in $U(x)$ can agree on their intersection. We therefore say that two elements $a \in \mathcal{S}(U_1)$ and $b \in \mathcal{S}(U_2)$ are equivalent if $r^1_{12}(a) = r^2_{12}(b)$. If we write the equivalence relation as $a \sim b$, then our desired object is given by

$$\mathcal{S}_x = \left(\bigsqcup_{U(x)} \mathcal{S}(U) \right) \Big/ \sim$$

and is called the *stalk* of $\mathcal{S}$ about x. We often call elements of a stalk a *germ.* Stalks can be used to generate a sheaf from a presheaf through a process called *sheafification* by gluing the stalks together in a way consistent with the sheaf axioms. The sheaf formed is then often called the *sheaf of germs* of the object in question.

A useful tool in topology is that of an *exact sequence* between topological spaces. We will also see that this notion extends to sheaves and will be an invaluable tool going forward, but to use these we must define what we mean by mapping between sheaves.

Definition 1.7. A *sheaf morphism* $f : \mathcal{S} \to \mathcal{T}$ is a collection of maps

$$f_U : \mathcal{S}(U) \to \mathcal{T}(U),$$

where f_U is a homomorphism of the objects contained within $\mathcal{S}(U)$ and $\mathcal{T}(U)$ that is compatable with the restriction homomorphism defined on both sheaves.

Definition 1.8. An *exact sequence of spaces* is a collection of spaces M_i and mappings $f_i : M_{i-1} \to M_i$, denoted by

$$M_0 \xrightarrow{f_1} M_1 \xrightarrow{f_2} \ldots \xrightarrow{f_{n-1}} M_{n-1} \xrightarrow{f_n} M_n,$$

such that the maps satisfy

$$\text{image}(f_k) = \ker(f_{k+1}),$$

for all $k = 1, \ldots, n-1$. Of particular interest is a *short exact sequence*, which has the form

$$0 \to M_1 \xrightarrow{f} M_2 \xrightarrow{g} M_3 \to 0.$$

Here 0 denotes the space containing only the identity element and since the maps into and out of 0 are completely determined we do not label them on in the sequence.

The definition of an exact sequence of sheaves is, in essence, the same as that for the case of spaces but with the understanding that we are always considering sheaf morphisms which are collections of maps. An interesting, but equivalent, statement that highlights the locality of exactness is given below.

Proposition 1.3. A sequence of sheaves $\mathcal{F} \xrightarrow{a} \mathcal{G} \xrightarrow{b} \mathcal{H}$ over a topological space M is ***exact*** if and only if the induced sequence on the stalks of the sheaves

$$\mathcal{F}_x \xrightarrow{a_x} \mathcal{G}_x \xrightarrow{b_x} \mathcal{H}_x$$

is exact for all $x \in M$.

Example 1.6. The most clear example of an exact sequence of sheaves is the ***exponentiation sequence.*** Define the following two sheafs over a complex manifold $\mathcal{M}$:

$$\mathcal{Z} = \{\mathcal{Z}(U) = \text{constant functions } z : U \to \mathbb{Z}\}$$
$$\mathcal{O}^* = \{\mathcal{O}^*(U) = \text{non-vanishing holomorphic functions } f : U \to \mathbb{C} \text{ under multiplication}\}.$$

Then the sequence

$$0 \to \mathcal{Z} \xrightarrow{\iota_{\mathcal{O}}} \mathcal{O} \xrightarrow{\exp} \mathcal{O}^* \to 0$$

is exact if $\iota_{\mathcal{O}}$ is the inclusion map and

$$\exp(f) = e^{2\pi i f}.$$

To see this we note that we clearly have $\ker(\exp) \cong \mathcal{Z}$ and, due to the fact that on any simply connected neighbourhood we can take an appropriately chosen branch of a logarithm, exp is surjective. If we considered instead the same sequence but restricted to an open subset U we would find that this sequence is no longer always exact, as this well-defined choice of logarithm may not exist. To study the obstruction to this sequence being exact we can employ the methodology of cohomology, which we detail below.

1.5 Čech and sheaf cohomology

First of all, let M be a topological space, $\mathcal{S}$ a sheaf over M and $U = \{U_a\}$ be an open covering of M. We then define a ***p-simplex*** σ to be an ordered $(p+1)$-tuple of open sets in U. Furthermore, if we write $\sigma = (U_0, ..., U_p)$, then the ***support*** of σ is defined as

$$|\sigma| = U_0 \cap ... \cap U_p = U_{01...p}.$$

We now want to introduce the sheaf theory into this, in a manner which greatly generalises that theory of singular cohomology. To do so, we define a ***p-cochain*** with coefficients in the sheaf $\mathcal{S}$ to be a mapping $f : \sigma \mapsto \mathcal{S}(|\sigma|)$. For simplicity, we define

$$f_{i_0...i_p} \equiv f((U_{i_0}, ..., U_{i_p})) \in \mathcal{S}(U_{i_0...i_p}).$$

This showcases that a p-cochain is a family of sections of appropriate restrictions of the sheaf $\mathcal{S}$. Furthermore, we assume that in this notation a p-cochain is skew-symmetric in all of its indices, as in [11]. We finally note that we denote the abelian group formed under pointwise addition of p-cochains as $C^p(U, \mathcal{S})$.

Now that we have cochains, the only object left to create a theory of cohomology is a nilpotent operator of order 2. To this end, if we consider a $(p+1)$-simplex $\sigma = (U_0, ..., U_{p+1})$, then the *coboundary operator* δ_p acting upon a p-cochain f is defined by

$$\delta_p f_{i_0 \dots i_{p+1}} = (p+1)\rho_{[i_{p+1}} f_{i_0 \dots i_p]},$$

where $\rho_i = r^{|\sigma_i|}_{|\sigma|}$ and $\sigma_i = (U_0, .., U_{i-1}, U_{i+1}, ..., U_{p+1})$[6]. It is easy to verify that these operators satisfy $\delta_{p+1} \circ \delta_p = 0$. As such we can conclude that $\text{image}(\delta_{p-1}) \subset \ker(\delta_p)$. We denote each of these subsets by

$$Z^p(U, \mathcal{S}) = \ker(\delta_p)$$
$$B^p(U, \mathcal{S}) = \text{image}(\delta_{p-1})$$

and call elements of $Z^p(U, \mathcal{S})$ *p-cocycles* and elements of $B^p(U, \mathcal{S})$ *p-coboundaries*. Since the set of p-coboundaries is contained within the set of p-cochains, we can quotient out the p-coboundaries to form the group of p-cocycles modulo p-coboundaries. The result is a sheaf theoretic analogy of the pth-cohomology group, which we now state.

Definition 1.9. Let M be a topological space with an open covering U and a sheaf $\mathcal{S}$ over M. The pth Čech cohomology group with respect to U is defined by

$$\check{\text{H}}^p(U, \mathcal{S}) = Z^p(U, \mathcal{S}) / B^p(U, \mathcal{S}).$$

What we have done is essentially analagous to the process of deriving the theory of singular cohomology, except with an extra step taking our simplices to a sheaf. The only peculiarity that we now have is that our cohomology groups depend on the choice of open covering. As a covering is arbitrary and often chosen to simplify calculations, we wish to form another group from the Čech cohomology groups which do not depend on the choice of cover. This issue can be resolved by the notion of *refinement*.

Let $U = \{U_i\}$ and $V = \{V_j\}$ be open covers of a topological space M and let I and J be the indexing sets for U and V respectively. We say that U is a refinement of V is there exists a map

$$h : I \to J \quad \text{such that} \quad U_i \subset V_{h(i)},$$

for all $i \in I$. This is simply the statement that each open set in U is a subset of some subset in V. The map h then induces a map on the Čech cohomology groups

$$h^* : \check{\text{H}}^p(U, \mathcal{S}) \to \check{\text{H}}^p(V, \mathcal{S}).$$

These induced maps can be used to set up an equivalence relation between Čech cohomology groups with respect to different covers. Suppose that $W = \{W_k\}$ is a common refinement of

[6] Technically speaking, using this index notation, we should be talking about the sets of these sections and the coboundary operator acting as a map into these sets, but we will abuse notation and leave out the brackets denoting the set.

both U and V with induced restriction maps h^* and l^* respectively. Then if $[f] \in \check{\mathrm{H}}^p(U,\mathcal{S})$ and $[g] \in \check{\mathrm{H}}^p(V,\mathcal{S})$ satisfy

$$h^*[f] = l^*[g] \in \check{\mathrm{H}}^p(W,\mathcal{S}),$$

we say that $[f]$ and $[g]$ are equivalent and denote it by $[f] \sim [g]$. Therefore, the cover independent group we require is

$$\check{\mathrm{H}}(M,\mathcal{S}) = \left(\bigsqcup_U \check{\mathrm{H}}^p(U,\mathcal{S}) \right) \Big/ \sim,$$

where the union is taken over all open coverings U of M. This is called the pth-*sheaf cohomology group of the space M with coefficients in the sheaf $\mathcal{S}$*.

In terms of practicality, the construction via refinements of the sheaf cohomology groups is not of great use. It does, however, point towards a possible simplification that aids in their construction.

The process above is an example of a *direct limit* and as such gives us a homomorphism between Čech cohomology groups and the sheaf cohomology groups. It is fortunate that in many cases this homomorphism is in fact an isomorhpism for certain covers. If this property holds, we call the cover a *Leray cover*. Alas, it is not clear whether such a cover always exists but the following theorem gives us a method of seeing if a cover is, indeed, a Leray cover.

Theorem 1.1. (Leray) Let M be a topological space with an open covering U and a sheaf $\mathcal{S}$ over it. If the cover U satisfies

$$\check{\mathrm{H}}^p(|\sigma|,\mathcal{S}) = 0$$

for $p \in \mathbb{N}$ and all simplices σ, then U is a Leray cover. That is,

$$\check{\mathrm{H}}^p(U,\mathcal{S}) \cong \check{\mathrm{H}}^p(M,\mathcal{S})$$

for each p.

From here on out we will assume that each cover we use is a Leray cover instead of checking theorem 1.1 holds.

Example 1.7. We will calculate $\check{\mathrm{H}}^1(\mathbb{CP},\mathcal{O}(-1)) := \check{\mathrm{H}}^1(\mathbb{CP},\Gamma(\mathbb{CP},\mathcal{O}(-1)))$ explicitly to showcase some of the notions defined above. We will cover $\mathbb{CP}$ by the two open sets

$$U_0 = \{[\pi_{0'},\pi_{1'}] \,||\, \pi_{0'} \neq 0\}$$
$$U_1 = \{[\pi_{0'},\pi_{1'}] \,||\, \pi_{1'} \neq 0\}$$

where $[\cdot,\cdot]$ refers to the equivalence of scaling in $\mathbb{CP}$. By example 1.3, the group $\check{\mathrm{H}}^1(\mathbb{CP},\mathcal{O}(-1))$ can be viewed as the space of functions that are holomorphic on $U_0 \cap U_1$ and are homogeneous of degree -1 modulo coboundaries. Suppose we have such a 1-cocycle F_{01}, then we can define the 0-cocycles

$$G_j(\pi_{A'}) = \frac{1}{2\pi i} \oint_{\Gamma_j} \frac{F_{01}(\zeta_{A'})}{\zeta^{B'}\pi_{B'}} \zeta_{C'}\, \mathrm{d}\zeta^{C'},$$

where the contours are shown in figure 2 and $j \in \{0,1\}$.

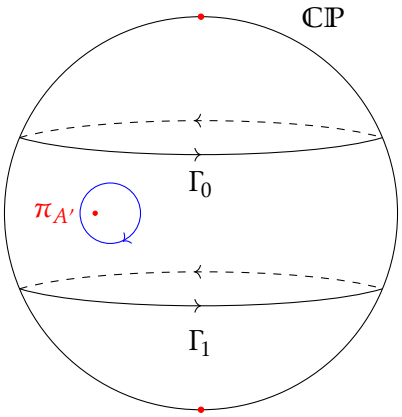


Figure 2: The contours integrated over in G_j. Also shown in blue is the contour $\Gamma = \Gamma_0 - \Gamma_1$.

We can expand the summed indices and use the homogeneity of F_{01} to reduce this to

$$G_j(\pi_{A'}) = \frac{1}{2\pi i} \oint_{\Gamma_j} \frac{F_{01}(\zeta, 1)}{\zeta \pi_{1'} - \pi_{0'}} \, \mathrm{d}\zeta,$$

where $\zeta = \frac{\zeta^{0'}}{\zeta^{1'}}$. The difference of the two 0-cocycles is therefore

$$G_0(\pi_{A'}) - G_1(\pi_{A'}) = \frac{1}{2\pi i} \oint_{\Gamma} \frac{F_{01}(\xi, 1)}{\xi \pi_{1'} - \pi_{0'}} \, \mathrm{d}\xi = F_{01}(\pi_{A'}),$$

by Cauchy's residue theorem. This exhibits that any 1-cocycle is in fact a 1-coboundary. We therefore conclude that $\check{\mathrm{H}}^1(\mathbb{CP}, \mathcal{O}(-1)) = 0$. By noting that dividing any homogeneous polynomial of degree n by some other polynomial homogeneous of degree $(n+1)$ gives us a function homogeneous of degree -1, we see that this case is equivalent to the calculation above. This in turn tells us that we have $\check{\mathrm{H}}^1(\mathbb{CP}, \mathcal{O}(n)) = 0$ for $n > 0$.

Finally, we end this section by noting an extremely useful result about the sheaf cohomology groups. The following theorem gives a powerful way of constructing exact sequences of cohomology groups and will see extensive use later. We will not prove it as the proof simply comes down to exhibiting a suitable diagram of sequences and complexes together with constructing a connecting map and hence refer readers to [10] for more details.

Theorem 1.2. Let M be a topological space with three sheaves $\mathcal{F}, \mathcal{G}$ and $\mathcal{H}$ over it. If the short sequence

$$0 \to \mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H} \to 0$$

is exact, then the long sequence of cohomology groups

$$0 \to \check{\mathrm{H}}^0(M, \mathcal{F}) \xrightarrow{\alpha^*} \check{\mathrm{H}}^0(M, \mathcal{G}) \xrightarrow{\beta^*} \check{\mathrm{H}}^0(M, \mathcal{H}) \xrightarrow{\delta^*} \check{\mathrm{H}}^1(M, \mathcal{F}) \to \ldots$$

is exact.

2 The twistor space

Now that we have looked over many of the mathematical prerequisites needed we task ourselves with constructing twistor space, which houses the fundamental objects of our theory. After discussing a familiar space from physics, Minkowski space, we will develop not only the twistor space but the correspondence between itself and a symmetrically completed version of Minkowski space, therefore giving us a basis upon which to study mathematical physics. We finally develop an equivalent picture which will shed some extra light on the geometry of twistors.

2.1 Compactified Minkowski space

A large portion of theoretical physics, including quantum field theory, typically takes place in Minkowski space $\mathbb{M}$ as its isometry group is the Poincaré group. If we endevour to extend the symmetries of this space to include conformal transformations we notice a peculiarity. To illustrate this, consider a *special conformal transformation* $x^a \mapsto y^a$ given by

$$y^a = \frac{x^a + x^2 b^a}{1 + 2(b \cdot x) + b^2 x^2}.$$

The issue that arises from this is that there are choices of b^a that lead to an infinite y^a. This tells us that the Killing vector field associated to special conformal transformations is *incomplete* when we consider Minkowski space. To this end, we require some form of symmetric extension of Minkowski space that can allow for this type of behaviour. We do this by *compactifying* $\mathbb{M}$ to include some extra structures while still maintaining the finite structure of $\mathbb{M}$.

A strikingly simple compactification of $\mathbb{M}$ can be derived by embedding Minkowski space into a quadric of $\mathbb{RP}^5$ and taking the closure of the space to gain the required extra structure. To do this we first consider the space $\mathbb{R}^6$ with coordinates (u, v, w, x, y, z) and the mixed signature metric

$$\mathrm{d}s^2 = \mathrm{d}u^2 + \mathrm{d}v^2 - \mathrm{d}w^2 - \mathrm{d}x^2 - \mathrm{d}y^2 - \mathrm{d}z^2.$$

With this choice of metric, the null cone at the origin $\mathbb{N}$ is defined by

$$u^2 + v^2 - w^2 - x^2 - y^2 - z^2 = 0.$$

This relation is invariant under scaling, so we can therefore consider only the projectified null cone $\mathbb{PN}$. The use of $\mathbb{PN}$ is vital, as this is a compact four dimensional manifold upon which we can embed Minkowski space into via the map

$$X^\mu \mapsto (X^0, \tfrac{1}{2}(1 - X^\mu X_\mu), -\tfrac{1}{2}(1 + X^\mu X_\mu), X^1, X^2, X^3). \tag{2.1}$$

Furthermore, it can be easily seen that this map acts as an embedding into the intersection of $\mathbb{PN}$ and the plane $v - w = 1$, therefore telling us that the embedding is in fact isometric. Through this map we can say that $\mathbb{PN} \subset \mathbb{RP}^5$ is a compactification of Minkowski space, which we denote by $\mathbb{M}^c$

Now that we have a compactification we want to find out what extra structure we have and if it remedies the problem arising from demanding conformal symmetry.

Any point in the non-projectified null cone $\mathbb{N}$ that satisfies $v - w \neq 0$ can be identified with a point in Minkowski space. This is due to the fact that we can scale this point such that $v' - w' = 1$ and use the map above to regain our point in $\mathbb{M}$. The points in $\mathbb{PN}$ that satisfy $v - w = 0$ cannot be scaled so that $v' - w' = 1$ and therefore do not belong to the image of Minkowski space. These are the extra points that we gain from the compactification and can be viewed as the intersection of $\mathbb{PN}$ with a null plane through the origin. All such planes are invariant under the group action of $O(2,4)$, the double cover of the conformal group in Minkowski space. Because this is the case, we have that the Killing vector field corresponding to special conformal transformations is complete on $\mathbb{M}^c$.

To investigate what these extra points represent in we define the following two subsets of $\mathbb{N}$

$$\mathbb{N}_1 = \{X \in \mathbb{N} |\ v = w \neq 0\},$$
$$\mathbb{N}_2 = \{X \in \mathbb{N} |\ v = w = 0\}$$

and consider their projectified versions. In $\mathbb{PN}_1$ we can choose a scaled representative of $X \in \mathbb{N}_1$ such that $v' = w' = 1$. It then follows by checking the norms of the vectors that $\mathbb{PN}_1$ has the structure of a null cone. This is the ***null cone at infinity*** that Huggett & Tod [11] claims that represents all additional points but, as noted by Jadczyk [12], there is an additional structure coming from $\mathbb{PN}_2$. Any element $X \in \mathbb{N}_2$ with $X = (u, 0, 0, \boldsymbol{x})$ can be scaled so we have $u' = 1$. By the nullness of this vector, we must have $|\boldsymbol{x}|^2 = 1$, telling us that $\mathbb{PN}_2$ has the structure of a 2-sphere. We therefore conclude that compactified Minkowski space extends regular Minkowski space by the addition of a null cone at infinity and an, often overlooked, additional 2-sphere.

2.2 Twistor space and the twistor correspondence

As a starting point towards constructing the twistor correspondence we wish to adopt a useful description that makes a correspondence between certain spaces clear. To do so, we define the following two manifolds.

Definition 2.1. The k-dimensional *Grassmannian manifold* of a vector space V is defined to be

$$G_k(V) = \{k\text{-dimensional subspaces of } V\}.$$

While we will not explicitly show that $G_k(V)$ is a manifold we do note that a possible coordinate chart for $G_k(\mathbb{C}^n)$ is given by

$$\chi(Z) = \left[\begin{pmatrix} iZ \\ \mathbb{I}_k \end{pmatrix}\right]$$

where Z is an $(n-k) \times k$ matrix, $\mathbb{I}_k$ is the $k \times k$ identity matrix and $[\cdot]$ takes the span of the columns of the matrix in question. The other charts required to cover the space can then be constructed by permuting the rows of image(χ).

Definition 2.2. Let V be a vector space of dimension n and let $\{a_1, ..., a_n\}$ be a sequence of positive integers satisfying $1 \leq a_1 < ... < a_m < n$. Then the $(a_1, ..., a_m)$-type ***flag manifold*** of V is defined to be

$$F_{a_1 ... a_m}(V) = \{(V_1, ..., V_m) |\ V_i \text{ are subspaces of } V \text{ of dimension } a_i,\ V_1 \subset V_2 \subset ... \subset V_m\}.$$

Notice that a flag manifold is essentially a generalisation of a Grassmannian manifold in the sense that if we take $m = 1$ in the definition above we recover the a_1-dimensional Grassmannian manifold of V.

Using the definitions above there is a natural mapping picture that can be exhibited by the diagram below

$$\begin{array}{ccc} & F_{12}(V) & \\ {\scriptstyle\mu}\swarrow & & \searrow{\scriptstyle\nu} \\ F_1(V) & & F_2(V) \end{array}$$

where μ is the natural projection onto $F_1(V)$ and ν is the natural projection onto $F_2(V)$. This is called a ***double fibration*** and this picture lies at the heart of the correspondence we will develop. A double fibration naturally gives us a way to associate subsets of the two projected upon spaces with each other through the map

$$\nu \circ \mu^{-1} : F_1(V) \to F_2(V)$$

and its compositional inverse $\mu \circ \nu^{-1} : F_2(V) \to F_1(V)$.

To obtain the double fibration picture that we require, we fix V to be a 4-complex dimensional vector space which we will call the ***twistor space*** $\mathbb{T}$. We denote the elements of twistor space by $Z^\alpha = (\omega^A, \pi_{A'})$ where α runs from 0 to 3 and both ω^A and $\pi_{A'}$ are spinors. It then follows that $F_2(\mathbb{T}) \cong F_2(\mathbb{C}^4) = G_2(\mathbb{C}^4)$ and $F_1(\mathbb{T}) \cong \mathbb{CP}^3$. Things can be tidied up a little more before we describe our completed picture; namely, it can be shown that $G_2(\mathbb{C}^4) \cong \mathbb{CM}^c$ (refer to Ward & Wells [10] for more details). Finally we arrive at the deseired double fibration:

$$\begin{array}{ccc} & F_{12}(\mathbb{T}) & \\ {\scriptstyle\mu}\swarrow & & \searrow{\scriptstyle\nu} \\ \mathbb{PT} & & \mathbb{CM}^c \end{array}$$

where we've denoted $\mathbb{CP}^3$ as $\mathbb{PT}$, the ***projective twistor space.*** In this context we call $F_{12}(\mathbb{T})$ the ***correspondence space*** and from here on denote it by $\mathbb{F}$.

Now that we have the correspondence between spaces, we wish to find a local correspondence by introducing coordinate charts to each space. This is most easily done by assigning coordinates to each space except at the "points at infinity". By this we mean, we give coordinate charts for the spaces given by

$$\boldsymbol{M^I} = \mathbb{CM},\ \boldsymbol{F^I} = \nu^{-1}(\boldsymbol{M^I}) \ \text{ and } \ \boldsymbol{P^I} = \mu \circ \nu^{-1}(\boldsymbol{M^I}).$$

The coordinates of $\boldsymbol{P^I}$ are simply given by a pair of homogeneous spinors $[\zeta^A, \pi_{A'}]$, while the coordinates of $\boldsymbol{M^I}$ is given by $x^{AA'} = x^a \sigma_a{}^{AA'}$ for $x^a \in \mathbb{CM}$. Coordinates for $\boldsymbol{F^I}$ then become obvious when we note that $\boldsymbol{F^I} \cong \boldsymbol{M^I} \times \mathbb{CP}$ as a consequence of its definition. We therefore arrive at the double fibration in terms of coordinates

$$\begin{array}{ccc} & (x^{AA'}, [\pi_{A'}]) & \\ {\scriptstyle\mu}\swarrow & & \searrow{\scriptstyle\nu} \\ [ix^{AA'}\pi_{A'}, \pi_{A'}] & & x^{AA'} \end{array}$$

Using these coordinates we notice that a twistor $Z^\alpha = (\zeta^A, \pi_{A'})$ is incident to a point $x^{AA'}$ in $\mathbb{CM}$ if and only if $\zeta^A = ix^{AA'}\pi_{A'}$. We call this the ***incidence relation*** and from it we can see, by fixing $x^{AA'}$, that a point in complexified Minkowski space corresponds to a line in projective twistor space. In the other direction we notice that a point in $\mathbb{PT}$ corresponds to a plane in $\mathbb{CM}$. We call these planes ***α-planes*** and it can be easily shown that they are totally self dual in the sense that every tangent bivector is self dual. These points are summaried in figure 3.

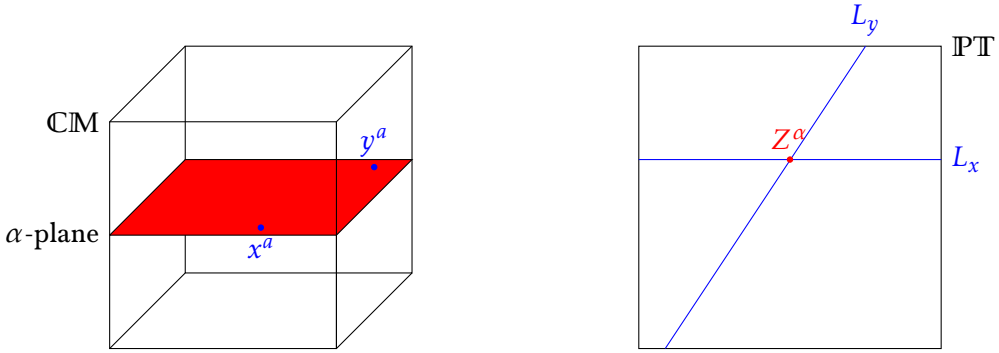


Figure 3: A pictorial representation of the twistor correspondence. The α-plane in red corresponds to the point Z^α in $\mathbb{PT}$ while the points x^a and y^a in $\mathbb{CM}^c$ correspond to projective lines L_x and L_y respectively. Since the two points lie on the same α-plane, the two lines L_x and L_y intersect at Z^α.

2.3 The Robinson congruence

The double fibration picture above is perhaps the most useful way in formulating the twistor correspondence due to its simplicity, but it has a drawback in the fact that it uses $\mathbb{CM}^c$ instead of real Minkowski space. We will therefore briefly explore an alternative formulation, due to Penrose [1], in which real Minkowski space takes a more prominent role and in fact leads to the reasoning behind the name twistor.

Our starting point this time is through the incidence relation $\omega^A = ix^{AA'}\pi_{A'}$, this time taking this as a fundamental property of twistors instead of a consequence of the double fibration. Twistors are then defined, in the same manner as before, as a pair of spinors $Z^\alpha = (\omega^A, \pi_{A'})$ with twistor space $\mathbb{T}$ now being the collection of all these pairs. In order to form a useful goemetric analysis, while still resorting only to real Minkowski space, we introduce the Hermitian form

$$\Sigma(Z, \bar{Z}) = Z^0\bar{Z}^2 + Z^1\bar{Z}^3 + \bar{Z}^0Z^2 + \bar{Z}^1Z^3 = \omega^A\bar{\pi}_A + \bar{\omega}^{A'}\pi_{A'}.$$

We now divide $\mathbb{T}$ into the three following subsets

$$\begin{aligned}
\mathbb{T}^+ &= \{Z \in \mathbb{T} |\ \Sigma(Z, \bar{Z}) > 0\}, \\
\mathcal{N} &= \{Z \in \mathbb{T} |\ \Sigma(Z, \bar{Z}) = 0\}, \\
\mathbb{T}^- &= \{Z \in \mathbb{T} |\ \Sigma(Z, \bar{Z}) < 0\}.
\end{aligned}$$

The reasoning behind this partitioning of twistor space is that the α-plane and projective line argument that applied on $\mathbb{CM}$ does not extend to real Minkowski space. Indeed, we will have to consider two cases; null projective twistors, that belong to $\mathbb{PN}$, and non-null projective twistors, that belong to either $\mathbb{PT}^+$ or $\mathbb{PT}^-$.

Fixing x^a in the incidence relation tells us that it corresponds to a line in $\mathbb{PT}$, as before. Furthermore, this line must lie within $\mathbb{PN}$ as a result of requiring x^a to be real. Conversely, fixing the twistor in the incidence relation to lie within $\mathbb{PN}$ gives rise to a null ray in $\mathbb{M}$.

While the case of null twistors is somewhat similar to the cases we derived using $\mathbb{CM}$, non-null twistors are trickier to interpret in $\mathbb{M}$. To do so, we must note that the Hermitian form Σ defines a space $\mathbb{T}^*$ which is dual to $\mathbb{T}$. So for any twistor $Z^\alpha \in \mathbb{PT}^+ \cup \mathbb{PT}^-$, there exists a dual element $\bar{Z}^\alpha \in \mathbb{PT}^*$. This dual element corresponds to a complex plane $\mathbb{CP}^2$ in $\mathbb{PT}$. As this plane has real dimension 4 and $\mathbb{PN} \cong S^2 \times \mathbb{R}^3$ has real dimension 5, we conclude that the plane corresponding to $\bar{Z}^\alpha$ intersects $\mathbb{PN}$ in a 3 dimensional locus. This 3 dimensional locus corresponds to a 3 parameter set of null rays in Minkowski space that we call the *Robinson congruence*. We therefore identify each non-null twistor with the Robinson congruence generated by its dual twistor. Figure 4 gives a simple example of this and shows the "twisting" of null rays represented by the congruence. It is for this exact reasoning that twistors are named as such.

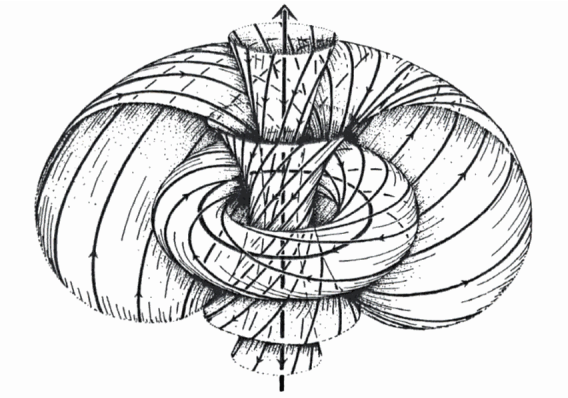


Figure 4: Visualisation of the Robinson congruence taken from Penrose and Rindler. [13]

3 The zero rest mass equations

Having looked over the mathematical prerequisites needed, we will apply twistor theory to the simplest, yet still intuitive, system that we can- the *zero rest mass (ZRM) equations*. The ZRM equations have a natural physical interpretation that we will first explore and then apply our twistor methods to find solutions of it. In the final part of this section we will also see sheaf cohomology play a major role and will work through the details somewhat carefully so we can build more understanding on the use of sheaves.

3.1 Physical motivation

Free particles in quantum field theory are described by a relativistic wave equation. Indeed, if a particle is described by a field φ, then it satisfies

$$(\Box + m^2)\varphi = 0,$$

where $\Box = \nabla^a \nabla_a$ is the *d'Alembertian operator*. This is the most general relativistic wave equation for particles described by scalar fields, but for other representations of the Lorentz group this is no longer the most general equation. One only needs to look at spin-$\frac{1}{2}$ particles to find that a spinor ψ obeys

$$(i\gamma^a \nabla_a - m)\psi = 0,$$

the famous *Dirac equation*. We could continue to list these free particle equations for higher spin particles but it would be more convenient if we could write down an equation that describes a free particle described by a field from an arbitrary representation of the Lorentz group. This is exactly what Bargmann and Wigner [14] set out to accomplish from a purely group theoretic stand point. Specialising to the case of massless particles, the result of this approach gives

$$\nabla_A{}^{A'} \varphi_{A'B'\ldots C'} = 0 \quad \text{and} \quad \nabla_{A'}{}^{A} \varphi_{AB\ldots C} = 0, \tag{3.1}$$

the ZRM equations, often called the *Bargmann-Wigner equations*. Before we justify why this indeed describes massless particles, we will briefly discuss the terms that appear in this equation.

The most familiar item in equation 3.1 is $\varphi_{A'B'\ldots C'}$, which, as discussed in section 1.3, we view as a section of the bundle $\pi : \bar{\mathbb{S}}^* \otimes \ldots \otimes \bar{\mathbb{S}}^* \to \mathbb{CM}$ while $\varphi_{AB\ldots C}$ is a section of the bundle conjugate to this. A particle described by a primed n-spinor is said to be of helicity $\frac{n}{2}$. If the particle is instead described by an unprimed n-spinor then we say that the particle has helicity $-\frac{n}{2}$.

In the ZRM equations we now have a new type of operator appearing; the *spinor covariant derivative* $\nabla_{AA'}$ that corresponds to the Levi-Civita connection in Minkowski space. Schematically, the Levi-Civita connection is extended to this case by first generalising to complex spaces in the natural way and then demanding the Leibniz property holds for arbitrary valence spinors. This is done in such a manner that $\nabla_{AA'} \epsilon_{BC} = 0$, which follows from computation after completing the previous step. This extension is unique in the sense that it is the only way to generalise the Levi-Civita connection while maintaining that the symplectic form is covariantly constant, much in same manner that the Levi-Civita connection is the unique torsion free connection that covariantly preserves the metric. For more details of uniqueness and construction, consult Penrose & Rindler [13].

To justify why the ZRM equations indeed generalise the free relativistic wave equation for massless particles, it suffices to show that they imply the Klein-Gordon equation with $m = 0$. This is easily seen by noting that $\nabla_{AB'}\nabla^{AA'} = \frac{1}{2}\delta_{B'}^{\ A'}\Box$. So applying $2\nabla_{D'}^{\ A}$ to the ZRM equations gives us

$$\delta_{D'}^{\ A'}\Box\varphi_{A'B'\ldots C'} = \Box\varphi_{D'B'\ldots C'} = 0,$$

which verifies our claim that $\varphi_{A'\ldots B'}$ represents a massless field.

3.2 Contour integral solution

As expected of any free particles equation of motion, the ZRM equations can be solved exactly in a closed form solution. Penrose [15] gave the solution as a contour integral, which in modern language can be written as

$$\varphi_{A'\ldots B'} = \frac{1}{2\pi i}\oint_\Gamma \pi_{A'}\ldots\pi_{B'}\rho_x f(Z^\alpha)\pi_{C'}\,\mathrm{d}\pi^{C'}, \tag{3.2}$$

where the symbol ρ_x denotes the restriction to the line L_x corresponding to the spacetime point x. It is also worth noting that here f is called a ***twistor function*** and the contour Γ is arbitrary given that it does not include any singularities of f. For the integrand to be well defined on $\mathbb{CP}$ we notice that f must be homogeneous of degree $(-n-2)$ if $\varphi_{A'\ldots B'}$ has helicity $\frac{n}{2}$.

The fact that 3.2 solves the ZRM equations can be easily shown. Indeed, we only need to notice that

$$\frac{\partial}{\partial x^{AA'}}\rho_x f(Z^\alpha) = i\pi_{A'}\rho_x\left(\frac{\partial f}{\partial\omega^A}\right),$$

for a twistor $Z^\alpha = (\omega^A, \pi_{A'})$. It then follows that

$$\nabla_A^{\ D'}\varphi_{A'\ldots B'} = \frac{1}{2\pi}\oint_\Gamma \pi_{A'}\pi^{D'}\ldots\pi_{B'}\rho_x\left(\frac{\partial f}{\partial\omega^A}\right)\pi_{C'}\,\mathrm{d}\pi^{C'},$$

which upon contraction of A' and D', gives zero. The case of a scalar particle is a special case, as the resulting equation of motion is second order, but the principle still applies. Indeed, after two derivatives we get

$$\frac{\partial^2\varphi}{\partial x^{AA'}\partial x^{BB'}} = \frac{i}{2\pi}\oint_\Gamma \pi_{A'}\pi_{B'}\rho_x\left(\frac{\partial^2 f}{\partial\omega^A\partial\omega^B}\right)\pi_{C'}\partial\pi^{C'},$$

which, again, gives zero after contraction. Simalarily, we can write down a solution for the unprimed ZRM equation as

$$\varphi_{A..B} = \frac{1}{2\pi i}\oint_\Gamma \rho_x\left(\frac{\partial}{\partial\omega^A}\ldots\frac{\partial}{\partial\omega^B}f(Z^\alpha)\right)\pi_{C'}\,\mathrm{d}\pi^{C'}. \tag{3.3}$$

It is clear from the arguments above that this is, as claimed, a solution to the ZRM equations.

The form of the solutions 3.2 and 3.3 should not be of much surprise as it bears resemblance to standard integral transforms such as a Fourier transform or, perhaps more appropriately

given the dependence on contours, a generalised Poisson transform. It can be thought of as taking some arbitrary twistor function and transforming it into a solution to the ZRM equations, but two questions are raised simply by inspecting the form of the solutions.

The first question we ask is whether the contour Γ always exists. This is a subtle point and we will return to it later. The more pertinent question, however, is given a ZRM field, is there a unique twistor function that gives rise to it? If not, can we obtain the set of functions that give rise to it? We can answer the first part of this question quite easily.

Suppose we have a solution to the ZRM equations that has the form of solution 3.2 with the twistor function f generating it. Now represent L_x as the Riemann sphere and form the closed contour Γ on the surface of the sphere. We then use this contour to split the Riemann sphere into two patches D_L and D_R, as in figure 5.

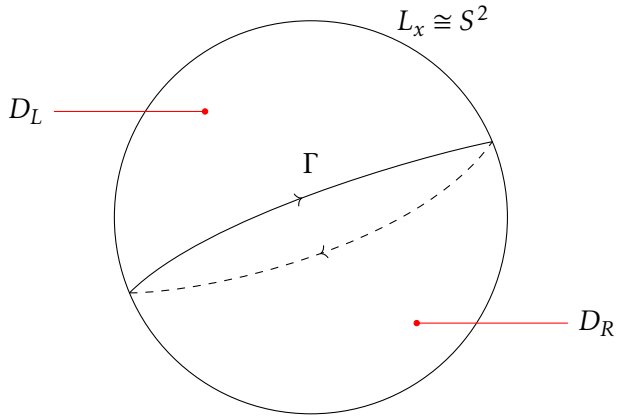


Figure 5: A closed contour Γ can be used to split the Riemann sphere into two distinct patches D_L and D_R.

Now define h_L to be a holomorphic function on D_L and h_R to be holomorphic on D_R. It is then clear that the function $f + h_L - h_R$ gives rise to the same ZRM field. So now we know that each ZRM field doesn't have a unique twistor function representative, but it is not clear if we can, in essence, invert the transform or not. In order to tackle this problem we will need to use some of the sheaf theory that we established earlier.

3.3 The Penrose transform

Recall that given an open covering of $\mathbb{CP}$, say (U_0, U_1), the group $\check{H}^1(\mathbb{CP}, \mathcal{O}(n))$ can be viewed as the space of functions holomorphic on $U_0 \cap U_1$ homogeneous of degree n modulo coboundaries. In essence, this is very similar to what we described above; we implicitly formed an open covering of $\mathbb{CP}$ and found that adding functions holomorphic on either patch to a given twistor function gives no change. These, in fact, correspond to coboundaries in the sheaf theory picture. This motivates us to flesh out this approach and use sheaves to illuminate the relation between ZRM fields and twistor functions. We will do so by following arguments similar to those found in chapter 10 of Huggett and Tod [11].

The natural method to identify this relation is to set up a short exact sequence of sheaves and then look at the corresponding long exact sequence of cohomology groups. As we want

to relate objects on $\mathbb{CM}^c$ to those on $\mathbb{PT}$, we make the decision to look at sheaves over the correspondence space $\mathbb{F}$ and then aim to push down the results to the spaces we want. Crucially, a function f defined on $\mathbb{F}$ is also well defined on $\mathbb{PT}$ if it is constant on α-planes; or equivalently

$$\pi^{A'}\nabla_{AA'}f = 0, \tag{3.4}$$

while f is well defined on $\mathbb{CM}^c$ if it is independent of the spinor $\pi_{A'}$. However, it is more useful to restrict our attention to the most relevant subset of $\mathbb{CM}^c$ called the *future tube* $\mathbb{CM}^+ = \nu \circ \mu^{-1}(\mathbb{PT}^+)$. We choose to restrict to the future tube due to the fact that a field on $\mathbb{M}$ can be analytically extended to $\mathbb{CM}^+$ if and only if it is a positive frequency field [16], which are physically more relevent. We therefore have to look at sheaves over the space $\mathbb{F}^+ = \mu^{-1}(\mathbb{PT}^+)$ instead of over the entire correspondence space.

Now define the sheaf $\mathcal{Z}'_n(m)$ to be the sheaf consisting of germs of symmetric n-index primed spinors homogeneous of degree m in $\pi_{A'}$ that also satisfy the primed ZRM equation and are holomorphic on $\mathbb{F}^+$. A simple sheaf morphism that we will use in our sequence is given by

$$\Xi : \mathcal{Z}'_{n+1}(m) \to \mathcal{Z}'_n(m+1), \qquad \Xi : \varphi_{A'B'\dots C'} \mapsto \pi^{A'}\varphi_{A'B'\dots C'}.$$

This map can easily be shown to be surjective. Let $\phi_{B'\dots C'} \in \mathcal{Z}'_n(m+1)$ and define $\varphi_{A'B'\dots C'} \in \mathcal{Z}'_{n+1}(m)$ by

$$\varphi_{i'B'\dots C'} = \frac{1}{2\pi^{i'}}\phi_{B'\dots C'},$$

where $i' \in \{1', 2'\}$. Since $\mathbb{F}^+$ doesn't contain the zero spinor by definition, the only subtlety is when one of $\pi^{0'}$ or $\pi^{1'}$ is zero. This is easily remedied, however. If $\pi^{1'}$ is zero, for example, then by instead taking $\varphi_{1'B'\dots C'} = \lambda\phi_{B'\dots C'}$ for any constant $\lambda \in \mathbb{C}$ and doubling the definition above for $\varphi_{0'B'\dots C'}$ gives us that $\Xi(\varphi) = \phi_{B'\dots C'}$. It therefore follows that in any case $\Xi(\varphi) = \phi_{B'\dots C'}$ on some neighbourhood around any point; verifying our claim of surjectivity.

Notice now that if $\psi_{A'\dots B'} \in \ker(\Xi)$ then we must have $\psi_{A'\dots B'} = \pi_{A'}\dots\pi_{B'}f(x,\pi)$ where $f(x,\pi)$ is a twistor function of homogeneity $(-n-2)$ in $\pi_{A'}$. This is a simple consequence of $\varphi_{A'\dots B'}$ being symmetric and $\pi^{A'}\pi_{A'} = 0$. Furthermore, the ZRM equations require that this twistor function obeys 3.4. So if we let $\mathcal{T}(-n-2)$ denote the sheaf of germs of all such twistor functions, then the sequence

$$0 \to \mathcal{T}(-n-2) \xrightarrow{\zeta} \mathcal{Z}'_{n+1}(-1) \xrightarrow{\Xi} \mathcal{Z}'_n(0) \to 0, \tag{3.5}$$

is exact if the map ζ is given by

$$\zeta : f(x,\pi) \mapsto \underbrace{\pi_{A'}\pi_{B'}\dots\pi_{C'}}_{(n+1)\text{ many}} f(x,\pi).$$

As we have an exact sequence we can, with the use of theorem 1.2, form a long exact sequence of cohomology groups. The most relevant part of the resulting sequence is

$$\dots \to \check{\mathrm{H}}^0(\mathbb{F}^+, \mathcal{Z}'_{n+1}(-1)) \to \check{\mathrm{H}}^0(\mathbb{F}^+, \mathcal{Z}'_n(0)) \to \check{\mathrm{H}}^1(\mathbb{F}^+, \mathcal{T}(-n-2)) \to \check{\mathrm{H}}^1(\mathbb{F}^+, \mathcal{Z}'_{n+1}(-1)) \to \dots$$

The group $\check{H}^0(\mathbb{F}^+, \mathcal{Z}'_{n+1}(-1))$ is the group of global sections of the sheaf $\mathcal{Z}'_{n+1}(-1)$ over $\mathbb{F}^+$. If we fix x^a then these sections are regarded as global sections of homogeneity -1 over the projective line $L_x \cong \mathbb{CP}$; none of which exist. Therefore $\check{H}^0(\mathbb{F}^+, \mathcal{Z}'_{n+1}(-1)) = 0$. A harder, but similar, calculation shows that $\check{H}^1(F^+, \mathcal{Z}'_{n+1}(-1)) = 0$ as well[7]. We are therefore left with the isomorphism through the connecting map δ^* that reads

$$\check{H}^0(\mathbb{F}^+, \mathcal{Z}'_n(0)) \cong \check{H}^1(\mathbb{F}^+, \mathcal{T}(-n-2)). \tag{3.6}$$

This is the result that we want, albeit over $\mathbb{F}^+$ instead of the spaces we want. The final step is to examine these groups and push them down to twistor and Minkowski space.

An element of the left hand side of 3.6 is a function that satisfies the ZRM equations and is homogeneous of degree 0 in $\pi_{A'}$. It must therefore be independent of $\pi_{A'}$ and, as noted before, is in fact a function on $\mathbb{CM}^+$. So $\check{H}^0(\mathbb{F}^+, \mathcal{Z}'_n(0))$ can be regarded as the space of ZRM fields over complexified Minkowski space. On the other hand, an element of the right hand side of 3.6 is a twistor function of homogeneity $(-n-2)$ that obeys 3.4. As a consequence we can push this down to $\mathbb{PT}^+$ and note that we can replace $\mathcal{T}(-n-2)$ by the more familiar sheaf $\mathcal{O}(-n-2) = \Gamma(L^{-n-2})$. With these statements, the isomorphism 3.6 reads

$$\check{H}^1(\mathbb{PT}^+, \mathcal{O}(-n-2)) \cong \{\text{holomorphic ZRM fields } \varphi_{A'\dots B'} \text{ of helicity } \tfrac{n}{2} \text{ over } \mathbb{CM}^+\}. \tag{3.7}$$

Having dealt with positive helicity fields, can we do the same with negative helicity fields? The answer is yes, but the argument is slightly different so we will work through it briefly. First of all, let $\mathcal{K}(n)$ denote the sheaf of germs homogeneous of degree n in $\pi_{A'}$ that are holomorphic on $\mathbb{F}^+$. Now suppose $\psi \in \mathcal{K}(n)$. Define the mapping D_A by its action

$$D_A\psi = \pi^{A'}\nabla_{AA'}\psi.$$

Observe that if $\psi \in \ker(D_A)$ then ψ is homogeneous of degree n in $\pi_{A'}$ and pushes down to a twistor function. We therefore have the identification $\ker(D_A) \cong \mathcal{T}(n)$. If we define $\mathcal{Q}_A(n)$ to be the sheaf of germs of functions φ_A homogeneous of degree n in $\pi_{A'}$ that satisfy $D^A\varphi_A = 0$, then it is easy to verify that the sequence

$$0 \to \mathcal{T}(n) \xrightarrow{\iota} \mathcal{K}(n) \xrightarrow{D_A} \mathcal{Q}_A(n+1) \to 0$$

is exact. The only non-trivial part to this statement is the surjectivity of D_A, but this essentially follows from a modification of the Poincaré lemma so we will not dwell on it.

Proceeding as before, we have a corresponding long exact sequence of cohomology groups, the most relevant section of which is

$$\begin{aligned} 0 \to \check{H}^0(\mathbb{F}^+, \mathcal{T}(n)) \to \check{H}^0(\mathbb{F}^+, \mathcal{K}(n)) \to \check{H}^0(\mathbb{F}^+, \mathcal{Q}_A(n+1)) \\ \to \check{H}^1(\mathbb{F}^+, \mathcal{T}(n)) \to \check{H}^1(\mathbb{F}^+, \mathcal{K}(n)) \to \dots \end{aligned} \tag{3.8}$$

Once again, we must interpret these groups:

[7] The calculation relies on the fact that $\mathbb{CM}^+$ is a *Stein manifold* and that $\mathcal{Z}'_{n+1}(-1)$ is a *coherent analytic sheaf*, concepts which we will not readily need. We therefore refer readers to Grauert & Remmert [17] for a comprehensive primer on these topics. Of particular interest is Cartan-Serre theorem B, which is readily discussed in the text mentioned.

- The group $\check{H}^0(\mathbb{F}^+, \mathcal{T}(n))$ is the space of global twistor functions defined on $\mathbb{PT}^+$ that are homogeneous of degree n. If $\mu \in \check{H}^0(\mathbb{F}^+, \mathcal{T}(n))$, then μ is necessarily a polynomial in $\pi_{A'}$. We write this as

$$\mu = \mu_{A'\dots B'}\pi^{A'}\dots\pi^{B'}.$$

  As a consequence of the push down condition, these polynomials must additionally obey

$$\nabla_{C(C'}\mu_{A'\dots B')} = 0.$$

- $\check{H}^0(\mathbb{F}^+, \mathcal{K}(n))$ is the same as $\check{H}^0(\mathbb{F}^+, \mathcal{T}(n))$, but without the push down condition.
- $\check{H}^0(\mathbb{F}^+, \mathcal{Q}_A(n+1))$ consists of the spinor functions of the form $\psi_A = \psi_{AA'\dots C'}\pi^{A'}\dots\pi^{C'}$, where there are $(n+1)$ many prime index contractions, that also obey

$$D_A\psi^A = 0 \iff \nabla^A{}_{(D'}\psi_{A'\dots C')A} = 0.$$

- Similarly to the last case, we note that $\check{H}^1(\mathbb{F}^+, \mathcal{T}(n)) = \check{H}^1(\mathbb{PT}^+, \mathcal{O}(n))$.
- Finally, due to another argument based on Stein manifolds and coherent sheaves, we have

$$\check{H}^0(\mathbb{F}^+, \mathcal{Q}_A(n+1)) = 0.$$

We denote the first three of these groups by T_n, M_n and Ψ_n respectively. In this notation the exact sequence 3.8 becomes

$$0 \to T_n \xrightarrow{\iota} M_n \xrightarrow{\Theta} \Psi_n \xrightarrow{\delta^*} \check{H}^1(\mathbb{PT}^+, \mathcal{O}(n)) \to 0,$$

where δ^* is the connecting map, which we will not actually need to describe, and the map Θ is given by $\Theta : \mu_{B'\dots C'} \mapsto \nabla_{A(A'}\mu_{B'\dots C')}$.

Observe that if $\psi_A = \psi_{AA'\dots C'}\pi^{A'}\dots\pi^{C'} \in \Psi_n$, then we can define a ZRM field of helicity $-\frac{1}{2}(n+2)$ through the map

$$\Delta : \psi_A \mapsto \nabla^{B'}{}_{(B}\dots\nabla^{D'}{}_{D}\psi_{A)B'\dots D'}.$$

It is easy to verify this fact by simply applying $\nabla^{AE'}$ and using the condition $D^A\psi_A = 0$. In this context we call ψ_A a ***Hertz potential*** of a ZRM field. So we in fact have a second exact sequence of the form

$$0 \to T_n \xrightarrow{\iota} M_n \xrightarrow{\Theta} \Psi_n \xrightarrow{\Delta} \Phi_{n+2} \to 0,$$

where Φ_{n+2} is the set of ZRM fields of helicity $-\frac{1}{2}(n+2)$. Upon comparison of the two sequences we finally deduce our desired result

$$\check{H}^1(\mathbb{PT}^+, \mathcal{O}(n-2)) \cong \{\text{holomorphic ZRM fields } \varphi_{A\dots B} \text{ of helicity } -\tfrac{n}{2} \text{ over } \mathbb{CM}^+\}.$$

It is this result, together with statement 3.7, that Penrose exhibited in two seminal papers [18, 19] in the late 1970's. Together they form what is often called the *Penrose transform.*

We have now constructed the isomorphisms required for ZRM particles of non-zero helicity but we have neglected the case of scalar particles. Unsurprisingly this case also has a cohomological interpretation encapsulated by the statement

$$\check{\mathrm{H}}^1(\mathbb{PT}^+, \mathcal{O}(-2)) \cong \{\text{holomorphic scalar ZRM fields } \phi \text{ over } \mathbb{CM}^+\}.$$

This is exactly what one would expect by interpolating the two previous results but requires a separate proof as we would need to deal with a second order operator. For proof of this statement we refer readers to section 6 of Eastwood et al. [20].

As a final remark, we note that what we have done above is not strictly rigorous. We have skipped some technical details about the comparison of sheaves over different spaces and the treatment of conformal infinity in $\mathbb{CM}^c$, but these matters are hardly of concern for the mathematical physicist. It should suffice to know that following the years after Penrose published these results that it was quickly reformulated into a more sophisticated and rigorous language through the use of techniques such as spectral sequences. For details of this, consult the paper by Eastwood et al. [20]. Furthermore, the Penrose transform was extended from acting on holomorphic fields to hyperfunctions, which are a much broader class of possible solutions, by the likes of Wells [21].

3.4 Recovering the contour integral solution from cohomology

The only loose end we wish to tie up to finish this chapter is the matter relating the contour solution to the cohomological argument above. Of course it was the contour solution that inspired us to look at sheaves in the first place, but we can in fact totally recover it by analysing $(\delta^*)^{-1}$, the inverse of the connecting map.

Our starting point is the commutative diagram of cochain groups

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathcal{C}^0(\mathcal{T}(-n-2)) & \xrightarrow{\zeta^c} & \mathcal{C}^0(\mathcal{Z}'_{n+1}(-1)) & \xrightarrow{\Xi^c} & \mathcal{C}^0(\mathcal{Z}'_n(0)) & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle\delta} & & \downarrow{\scriptstyle\delta} & & \downarrow{\scriptstyle\delta} & & \\
0 & \longrightarrow & \mathcal{C}^1(\mathcal{T}(-n-2)) & \xrightarrow{\zeta^c} & \mathcal{C}^1(\mathcal{Z}'_{n+1}(-1)) & \xrightarrow{\Xi^c} & \mathcal{C}^1(\mathcal{Z}'_n(0)) & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle\delta} & & \downarrow{\scriptstyle\delta} & & \downarrow{\scriptstyle\delta} & &
\end{array}$$

where each group is understood to be over $\mathbb{F}^+$ and then appropriately pushed down to $\mathbb{CM}^+$ or $\mathbb{PT}^+$ as above. It is clear that each row is exact through the maps used in the exact sequence 3.5 and each row is a complex.

Each cohomology class of $\check{\mathrm{H}}^1(\mathbb{PT}^+, \mathcal{O}(-n-2))$ can be defined by a representative cocyle f_{ij} such that

$$\pi_{A'} \ldots \pi_{B'} \pi_{C'} f_{ij} \in \mathcal{C}^1(\mathcal{Z}'_{n+1}(-1)).$$

Evidently, applying δ to this gives zero and therefore defines a cohomology class of $\check{\mathrm{H}}^1(\mathbb{F}^+, \mathcal{Z}'_{n+1}(-1))$. But as we stated above, this group is trivial and as a consequence we must have

$$\pi_{A'} \ldots \pi_{B'} \pi_{C'} f_{ij} = \rho_{[i} \psi_{j]A' \ldots B'C'},$$

for some $\psi_{jA'\dots B'C'} \in \mathcal{C}^0(\mathcal{Z}'_{n+1}(-1))$. We can now apply Ξ^c, the cochain equivalent of the map Ξ, to this $\psi_{iA'\dots B'C'}$ to obtain

$$\varphi_{iA'\dots B'} := \Xi^c(\psi_{iA'\dots B'C'}) = \psi_{iA'\dots B'}\pi^{C'} \in \mathcal{C}^0(\mathcal{Z}'_n(0)).$$

We now make use of the isomorphism 3.6 to note that there must exist a ZRM field $\varphi_{A'\dots B'}$ that satisfies

$$\rho_j\varphi_{A'\dots B'} = \varphi_{jA'\dots B'}.$$

The final step is to split $\varphi_{jA'\dots B'}$. We know that $\pi_{A'}\dots\pi_{B'}\pi_{C'}\rho_x f_{ij}$ is a section of $\mathcal{O}(-1)$ over $\mathbb{CP}$ for a fixed x so, using our usual Leray patch for $\mathbb{CP}$, we can now use the argument we used in example 1.7 to obtain

$$\varphi_{jA'\dots B'} = \frac{\pi^{C'}}{2\pi i}\oint_{\Gamma_j} \zeta_{A'}\dots\zeta_{B'}\zeta_{C'}\frac{\rho_x f_{01}}{\zeta_{E'}\pi^{E'}}\zeta_{D'}\,\mathrm{d}\zeta^{D'},$$

which under relabelling the spinors and tidying up becomes

$$\varphi_{jA'\dots B'} = \frac{1}{2\pi i}\oint_{\Gamma_j} \pi_{A'}\dots\pi_{B'}\rho_x f_{01}(Z^\alpha)\pi_{C'}\,\mathrm{d}\pi^{A'},$$

as desired. This has not only replicated the result that we want, it also illuminates the issue of whether the contour always exists. Indeed, we can take the contours to be those as in example 1.7 and we will have no issue.

4 The Yang-Mills equations

The Yang-Mills equations have not only found extensive use in theoretical physics, but also have a natural geometric interpretation making them interesting to physicists and mathematicians alike. It was shown by Ward in 1977 that there is a correspondence between solutions to the anti-self dual sector and holomorphic vector bundles dubbed the *Penrose-Ward correspondence*. In this section we start by deriving the Yang-Mills equations from an action principle following the style of Figueroa-O'Farrill [22] by using the language of exterior forms and principle fibre bundles. We then aim to discuss the Penrose-Ward correspondence and highlight some of the applications of it.

4.1 The Yang-Mills action and equations

Let $\mathcal{M}$ be a complex manifold with a semisimple matrix gauge group G, a volume form ω and A a connection on said manifold. Furthermore, denote the adjoint bundle that fibres over $\mathcal{M}$ by $\mathrm{ad}P$. Then we define the gauge field (also called the field curvature) F to be

$$F = \mathrm{d}A + A \wedge A.$$

For us to build an action out of F we need to see how it transforms under the gauge transformation $A \mapsto gAg^{-1} - (\mathrm{d}g)g^{-1}$ for $g \in G$. It is simple to see that, after a bit of algebra, we get

$$F \mapsto gFg^{-1}.$$

This tells us that F transforms in the adjoint of G and can be viewed as an $\mathrm{ad}P$ valued 2-form on $\mathcal{M}$. The gauge field satisfies a *Bianchi identity* of the form

$$\mathrm{D}F = \mathrm{d}F + A \wedge F - F \wedge A = 0,$$

where D is the gauge covariant exterior derivative. The expression for D changes depending on the representation of the exterior form it is acting upon. For example, the expression for the components of D for a 1-form in the fundamental representation is given by $\mathrm{D}_i = \partial_i + A_i$ and from this expression we recover F via $F_{ij} = [\mathrm{D}_i, \mathrm{D}_j]$.

From these observations we can now see that[8]

$$S_{YM}[A] = \int_{\mathcal{M}} \mathrm{Tr}(F \wedge \star F) = \int_{\mathcal{M}} \langle F, F \rangle \omega$$

is a well defined gauge invariant action; we call this action the *Yang-Mills action*. Since G is taken to be semisimple, we note that the inner product taken above is induced by the Killing

[8] Strictly speaking there is nothing stopping us at this point from writing

$$\tilde{S}_{YM} = \int_{\mathcal{M}} \mathrm{Tr}(F \wedge F)$$

but this violates parity symmetry. This can be seen by noting that the parity operator reverses the orientation of the manifold, therefore picking up an overall minus sign in $\tilde{S}_{YM}$.

form, hence preserving gauge invariance. A connection is said to be a *Yang-Mills connection* if it minimises the action S_{YM}. This means that we require

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(S_{YM}[A+tB]\Big)\Big|_{t=0} = 0,$$

for any $B \in \Omega^1(\mathrm{ad}P)$. This can be evaluated simply by expanding the inner product within the action. Doing so gives us

$$0 = 2\int_{\mathcal{M}} \langle \mathrm{D}B, F\rangle\omega = 2\int_{\mathcal{M}} \langle B, \mathrm{D}^*F\rangle\omega.$$

It follows from our discussion of the Hodge star in proposition 1.2 that $\mathrm{D}^* = \star\mathrm{D}\star$ (up to a possible minus sign that we will neglect), so the Yang-Mills equations are therefore

$$\star\mathrm{D}\star F = 0 \iff \mathrm{D}\star F = 0, \tag{4.1}$$

where we have used the self duality of the Hodge star in the implication. From this we see that if one takes F to be either self dual or anti-self dual, the Bianchi identity automatically gives us the equations of motions and therefore tells us that the corresponding connection is a Yang-Mills connection. The *anti-self dual Yang-Mills equations* (ASDYM) are then simply

$$\star F = \begin{cases} -F & \text{in Riemannian signature} \\ -iF & \text{in Lorentzian signature} \end{cases}$$

The formulation of the Yang-Mills equations clearly have some level of underlying geometry to them, as one can infer from the above construction. Indeed, the formulation can be discussed very naturally in the language of fibre bundles. We further motivate this picture by noting that if we have a connection 1-form on the total space of the adjoint bundle, then the gauge potential A can be viewed as a pullback to the base space via local sections. This then extends to the field curvature 2-form, as this can be viewed as the pullback by local sections of the curvature 2-form on the total space. Gauge transformations then naturally arise as fibre preserving automorphisms.

This is obviously just a heuristic overview of the bundle-centric formulation and a more comprehensive overview, for those who require it, can be found in [22] and [23].

4.2 Lax pair and geometric interpretation

An important aspect of the ASDYM equations is that it admits a *Lax pair* formulation. To see this, we follow section 7 of Dunajski [4] and specialise to the case of $\mathcal{M} = \mathbb{CM} = \mathbb{C}^4$ with the following metric

$$\mathrm{d}s^2 = 2(\mathrm{d}z\,\mathrm{d}\tilde{z} - \mathrm{d}w\,\mathrm{d}\tilde{w}).$$

We can impose reality conditions on the coordinates to give various real flat metrics. In particular, if we choose

$$z = \frac{x^0 + ix^3}{\sqrt{2}}, \quad w = -\frac{x^1 + ix^2}{\sqrt{2}} \quad \text{with} \quad \tilde{z} = \bar{z} \text{ and } \tilde{w} = -\bar{w}$$

we recover Euclidean flat space, while taking

$$z = \frac{x^0 + x^3}{\sqrt{2}}, \quad w = -\frac{x^1 + ix^2}{\sqrt{2}} \quad \text{with} \quad \tilde{z} = \frac{x^0 - x^3}{\sqrt{2}} \quad \text{and} \quad \tilde{w} = \bar{w}$$

gives us the standard Minkowski spacetime. Furthermore, in these coordinates we take the volume form ω to be

$$\omega = \mathrm{d}w \wedge \mathrm{d}\tilde{w} \wedge \mathrm{d}z \wedge \mathrm{d}\tilde{z}.$$

With this set up it is not hard to see that the following 2-forms

$$\vartheta_1 = \mathrm{d}w \wedge \mathrm{d}z, \quad \vartheta_2 = \mathrm{d}w \wedge \mathrm{d}\tilde{w} - \mathrm{d}z \wedge \mathrm{d}\tilde{z} \quad \text{and} \quad \vartheta_3 = \mathrm{d}\tilde{w} \wedge \mathrm{d}\tilde{z}$$

are self dual and, in fact, form a basis for the space of self dual 2-forms. Since anti-self dual and self dual forms are orthogonal with respect to the wedge product, the ASDYM equations can be neatly written as $F \wedge \vartheta_i = 0$ for each i. This, in turn, tells us information about the components of F; namely that

$$F_{wz} = 0, \quad F_{w\tilde{w}} - F_{z\tilde{z}} = 0 \quad \text{and} \quad F_{\tilde{w}\tilde{z}} = 0.$$

It is this form of the ASDYM equations that lies at the heart of the Lax pair formulation.

Now consider the following situation. Take $\Theta \equiv \Theta(w, \tilde{w}, z, \tilde{z}; \zeta)$ to be the fundamental solution of the overdetermined system of equations

$$L\Theta(w, \tilde{w}, z, \tilde{z}; \zeta) = 0 \quad \text{and} \quad M\Theta(w, \tilde{w}, z, \tilde{z}; \zeta) = 0,$$

where L and M are linear operators and $\zeta \in \mathbb{CP}$ is an arbitrary parameter called a ***spectral parameter***. For the system to be consistent we must have the compatability condition $[L, M] = 0$ for all values of ζ. If we take L and M to be

$$L = \mathrm{D}_{\tilde{z}} - \zeta \, \mathrm{D}_w \quad \text{and} \quad M = \mathrm{D}_{\tilde{w}} - \zeta \, \mathrm{D}_z, \tag{4.2}$$

where in this case $\mathrm{D}_i = \partial_i + A_i$, we get that

$$[L, M] = F_{\tilde{z}\tilde{w}} - \zeta(F_{w\tilde{w}} - F_{z\tilde{z}}) + \zeta^2 F_{wz}.$$

The vanishing of this for all values of ζ therefore implies the ASDYM equations in the form we derived above. The operators L and M are said to be a Lax pair of the ASDYM system. The existence of a Lax pair depending on a spectral parameter is perhaps the defining property of integrable PDEs and many integrable systems admit a lax pair formulation[9]. This is the first hint that there is some level of integrability within the ASDYM equations.

The operators defined in equation 4.2 can be used to construct a geometric interpretation of the ASDYM equations. The first step towards this is to note that the vectors

$$l = \partial_{\tilde{z}} - \zeta \partial_w \quad \text{and} \quad m = \partial_{\tilde{w}} - \zeta \partial_z$$

[9]Examples of integrable systems that admit a Lax pair include the KdV equation, the non-linear Schrödinger equation and sine-Gordon equation.

span a totally null two-plane in $\mathbb{CM}^c$ and are related to the linear operators defined above by

$$L = l + l \lrcorner A \quad \text{and} \quad M = m + m \lrcorner A,$$

where the symbol $\lrcorner$ denotes the contraction of a vector with a p-form. For general null, and orthogonal l and m, these totally null two-planes come in two types as classified by the 2-form

$$\vartheta(\zeta) = \omega(l, m, -, -), \quad \vartheta_{\mu\nu} = l_{[\mu} m_{\nu]}.$$

By examining $\star\vartheta$ and its components, it is clear that it is annihilated by both l and m, telling us that $\star\vartheta \propto \vartheta$. Since the constant of proportionality must be an eigenvalue of the Hodge star, it tells us that ϑ must be either self dual or anti-self dual. For the vectors we've defined, we have

$$\vartheta(\zeta) = \vartheta_1 + \zeta\vartheta_2 + \zeta^2\vartheta_3,$$

for the self dual ϑ_i we defined above. This tells us that l and m define a totally null two-plane with self dual tangent bivector. This finally leads us to the main result.

Proposition 4.1. The ASDYM equations are equivalent to the statement that the field curvature $F = \mathrm{d}A + A \wedge A$ vanishes on each totally null self dual two-plane.

Proof. The ASDYM equations can be compactly written as $[L, M] = 0$ for the linear operators defined above. To use this fact we note that

$$F(l, m) = l^a m^b F_{ab} = [l^a \mathrm{D}_a, m^b \mathrm{D}_b].$$

Using the explicit forms of l and m, it is easy to see that

$$F(l, m) = [L, M] = 0.$$

After varying ζ, this is exactly the statement that the field curvature vanishes on null self dual two-planes. □

4.3 The Penrose-Ward correspondence

Proposition 4.1 gives us a natural geometric interpretation of the ASDYM equations that we can exploit to build up a twistoral theory of anti-self dual connections. In the following we will sketch the proof of the *Penrose-Ward correspondence*, leaving out some subtle technicalities and then discuss some useful results derived from it.

Theorem 4.1. (Ward [24]) There is a one-to-one correspondence between:

1. gauge equivalence classes of anti-self dual connections on $\mathbb{CM}$ with gauge group $\mathrm{GL}(n, \mathbb{C})$;
2. holomorphic rank-n vector bundles E over $\boldsymbol{P^I}$ which are trivial on each projective line L_x corresponding to a point $x \in \mathbb{CM}$.

We refer to this as the Penrose-Ward correspondence.

Proof. We choose to start by constructing the vector bundles and hence going from the first item to the second. Furthermore, we will do this abstractly, in a manner that doesn't highlight how to construct these bundles computationally, as we will discuss this later.

First of all, anti-self dual $\mathrm{GL}(n,\mathbb{C})$-gauge potentials can be viewed as sections of a bundle V whose fibres consist of the 1-forms that give rise to an anti-self dual curvature 2-form. Now let A be a connection on V, Z a point in $\boldsymbol{P^I}$ and P_Z the corresponding α-plane in $\mathbb{CM}$. This allows us to define the bundle E with the fibres

$$E_Z = \{\psi \in \Gamma(\mathbb{CM}, V) |\ v^a \mathrm{D}_a \psi = 0 \text{ for all } v^a \text{ tangent to } P_Z\}.$$

The condition $v^a \mathrm{D}_a \psi = 0$ is simply the statement that parallel transport around any loop leaves ψ unchanged. Since P_Z is simply connected, this is in fact equivalent to stating that the curvature must vanish on P_Z and is therefore equivalent to proposition 4.1. Since the curvature vanishes on α-planes, we clearly have that the parallel transportation condition uniquely defines ψ on P_Z in terms of some initial data; in this regard, we call $\mathrm{D} = \mathrm{d} + A$ *integrable*. As D is integrable, we must have that $E_Z \cong \mathbb{C}^n$ as the solutions are determined exactly by some initial data that takes a value in $\mathbb{C}^n$. Note that this whole process is holomorphic, so E must be a holomorphic vector bundle.

Does this bundle satisfy the triviality condition that we require? Well, let $\psi_x \in V_x$ be a 1-form at a point $x \in \mathbb{CM}$. This, by the integrability condition, defines a parallel transported field $\psi \in \Gamma(\mathbb{CM}, V)$ on each α-plane which in turn determines a point in E_Z for each Z that lies on L_x and therefore a section of $E|_{L_x}$. Now choosing n linearly independent 1-forms $\psi_x \in V_x$ gives us n linearly independent sections of the bundle E restricted to L_x, telling us that E is trivial when restricted to L_x.

On the other hand, consider a holomorphic rank-n $\mathrm{GL}(n,\mathbb{C})$-bundle E with the property that $E|_{L_x}$ is trivial for any $x \in \mathbb{CM}$. Then the space $\Gamma(L_x, E|_{L_x}) \cong \Gamma(\mathbb{CP}, \mathbb{CP} \times \mathbb{C}^n)$ is isomorphic to $\mathbb{C}^n$ by a version of Liouville's theorem. This space therefore endows each point x with a copy of $\mathbb{C}^n$, which we piece together to form a vector bundle V of rank-n over $\mathbb{CM}$. It is on this bundle that we want to form our connection.

A connection tells us how to compare two vectors living in two different spaces and therefore tells how to parallel transport objects along curves. If we describe how this is done, then we can construct our connection. To do so, note that we can pick a basis of $\mathbb{CM}$ in which all vectors are null. We therefore only need to examine transportation along null directions as we can form every direction from our null basis.

Suppose v^a is a null vector in $\mathbb{CM}$ that connects two points x and y. Then there exists a unique α-plane P_Z for which v^a is tangent to. Using the twistor correspondence (as in figure 3), we see that P_Z corresponds to a point Z in $\boldsymbol{P^I}$ through which the lines L_x and L_y pass through. Now given sections of $E|_{L_x}$ and $E|_{L_y}$ we can compare them by identifying them at the common point Z. This tells us how to transport from x to y and therefore defines a connection A on V which is flat on α-planes. Hence, by proposition 4.1, A is an anti-self dual Yang-Mills connection with gauge group $\mathrm{GL}(n,\mathbb{C})$. □

Remark. The proof exhibited is, as eluded to earlier, somewhat incomplete due to the omission of some minor details. We shall mention them here along with some additional information to paint a full picture of the theorem.

1. The most obvious omission is the fact that we did not prove that the connection in the second part was indeed a connection! This requires showing that it obeys certain axioms that are not mentioned in this report so we opt to leave it out.

2. We also chose to omit explicitly showing that the two constructions are mutually inverse, but this is simply a consequence of the geometry of the constructions and the proof of this amounts to little more than definition chasing.

3. Upon observing the proof one could ask if the result holds on some restriction of $\mathbb{CM}$. Indeed, the only topological argument above required the space $P_Z = \mathbb{CM} \cap P_Z$ to be simply connected. So if we demand that a subset $U \subset \mathbb{CM}$ obeys the condition that $U \cap P_Z$ is simply connected for all Z, then the Penrose-Ward correspondence holds over U the corresponding twistor space subset.

Having detailed the abstract proof, we will show how to explicitly, in principle, obtain the connection from a given bundle satisfying the triviality condition.

We choose to cover $\mathbb{PT}$ with two patches. Namely, we can take

$$U = \{(\omega^A, \pi_{A'}) |\ \pi_{0'} \neq 0\} \quad \text{and} \quad V = \{(\omega^A, \pi_{A'}) |\ \pi_{1'} \neq 0\}$$

to be our covering. Now let $F(Z^\alpha)$ denote the transition matrix holomorphic on $U \cap V$. We wish to impose the triviality condition now, so we let $\omega^A = ix^{AA'}\pi_{A'}$ in $F(Z^\alpha)$ to do so. Now by proposition 1.1, there must exists matrices H, holomorphic on $U \cap L_x$, and $\hat{H}$, holomorphic on $V \cap L_x$, that satisfy

$$F(ix^{AA'}\pi_{A'}, \pi_{A'}) = \hat{H}H^{-1},$$

on $U \cap V \cap L_x$. This splitting is the hardest part of the construction to impose and is often called the ***Riemann-Hilbert problem.*** Then, by the transition property of sections, each section of $E|_{L_x}$ is given by

$$s = H\psi,$$
$$\hat{s} = \hat{H}\psi$$

for some constant $\psi \in \mathbb{C}^n$ and for sections s and $\hat{s}$ defined on appropriate patches of $\mathbb{PT}$. Acting with $\pi^{A'}\nabla_{AA'}$ on $F(ix^{AA'}\pi_{A'}, \pi_{A'})$ clearly gives zero, giving us the condition

$$H^{-1}\pi^{A'}\nabla_{AA'}H = \hat{H}^{-1}\pi^{A'}\nabla_{AA'}\hat{H}.$$

Observe now that the left is holomorphic on $U \cap L_x$ while the right is holomorphic on $V \cap L_x$, implying that each expression is holomorphic on the whole of L_x. Furthermore, as both sides are homogeneous of degree 1 in $\pi^{A'}$, Liouville's theorem tells us that they must in fact be a polynomial of degree 1 in $\pi^{A'}$. We can therefore write the left hand side as

$$\pi^{B'}A_{BB'} = H^{-1}\pi^{A'}\nabla_{AA'}H,$$

for some A_b that is independent of $\pi^{A'}$. This is exactly the connection we want! But is it anti-self dual? Well, acting the relation above with $\pi^{C'}\nabla^B{}_{C'}$ gives us

$$\nabla^B{}_{(C'}A_{B')B} + A^B{}_{(C'}A_{B')B} = 0.$$

This relation is equivalent to the vanishing of the self-dual part of $A_{BB'}$, verifying that this is the required anti-self dual connection. We can, in principle, obtain all such connections by taking different coverings and hence different transition matrices $F(Z^\alpha)$. For simplicity, we need not specify the covering but just the twistor function $F(Z^\alpha)$, giving us a suitable way of generating each anti-self dual Yang-Mills connection.

With an eye towards describing physical fields, the Penrose-Ward correspondence may be considered somewhat unhelpful in its current form as it deals with $\mathrm{GL}(n,\mathbb{C})$ gauge fields as opposed to $\mathrm{SU}(n)$ gauge fields. Thankfully, we can reduce the gauge group from $\mathrm{GL}(n,\mathbb{C})$ to a more desirable group but this results in extra constraints upon the bundles over twistor space. Let us discuss the details of this briefly, highlighting the new structure required to describe $\mathrm{SU}(n)$ gauge fields.

First of all, in order to reduce the gauge group on the bundle V over $\mathbb{CM}$ we need to brandish the bundle with two extra structures. The first of which is that the fibres of V are given a covariantly constant volume form therefore reducing the group to $\mathrm{SL}(n,\mathbb{C})$. Secondly, as $\mathrm{SL}(n,\mathbb{C})$ is, in the sense of Lie groups, the real form of $\mathrm{SU}(n)$ we need to restrict our domain to a real subset which we choose to be S^4 for later convenience. Additionally, we need to give each fibre of V a positive-definite Hermitian form $\tau_0 : V \to V^*$ that is preserved by the connection on the bundle. By positive definite, we mean that for $\alpha, \beta \in V$ we have that $[\alpha,\beta] := \alpha(\tau_0\beta)$ is a well defined positive-definite inner product. This reduces our bundle V to an $\mathrm{SU}(n)$-bundle, but, as mentioned before, this also gives extra structure to E.

The reduction to $\mathrm{SL}(n,\mathbb{C})$ induces a holomorphically varying volume form on each fibre E_Z. The volume form is a nowhere vanishing global section of the line bundle $\det E = \bigwedge^n \mathrm{T}^*E$ which is trivial as a consequence[10].

In order to understand the effect of the reduction to $\mathrm{SU}(n)$ we must first define a notion of reality on $\mathbb{PT}$. To do so, we introduce the involution σ defined by

$$\sigma(Z^\alpha) = (\bar{Z}^1, -\bar{Z}^0, \bar{Z}^3, -\bar{Z}^2).$$

A line that connects the point Z to $\sigma(Z)$ is said to be *real line* and the space of real lines in $\mathbb{PT}$ is isomorphic to S^4. As real lines do not intersect we can say that $\mathbb{PT}$ is fibred by the real lines and this leads to the single fibration $\pi : \mathbb{PT} \to S^4$. We will assert without proof that this restriction to S^4 is compatible with the correspondence and refer readers to Ward & Wells [10] for more details. However, we want this involution to be lifted to E, so we use the Hermitian positive-definite form to do so. This means we have a map $\tau : E \to E^*$ that satisfies the following commutative diagram:

$$\begin{array}{ccc} E & \xrightarrow{\tau} & E^* \\ \downarrow & & \downarrow \\ \mathbb{PT} & \xrightarrow{\sigma} & \mathbb{PT} \end{array}$$

Furthermore, τ must be a *positive real form* in the sense that it satisfies the reality condition

$$[\zeta, \tau\rho] = \overline{[\rho, \tau\zeta]},$$

[10] A rank-n vector bundle is trivial if and only if there are n many global sections that form a basis on each fibre. Since a line bundle is of rank-1, the existence of a global section that vanishes nowhere guarantees triviality.

for all $\rho \in E_Z$ and $\zeta \in E_{\sigma(Z)}$. If one were to be rigourous, we would check that these arguements hold in both directions therefore verifying the one-to-one nature of the correspondence, but this is no more enlightening and we shall simply state our desired SU(n) correspondence.

Corollary 4.1. There is a one-to-one correspondence between:

1. gauge equivalence classes of anti-self dual connections on S^4 with gauge group SU(n);
2. holomorphic rank-n vector bundles E over $\mathbb{PT}$ such that $E|_{L_x}$ is trivial for all $x \in S^4$, the bundle $\det E$ is trivial and E admits a positive real form.

4.4 Yang-Mills instantons

It is well known, via Derrick's theorem for example, that static finite-energy solutions to the Yang-Mills equations exist in 4 dimensions and therefore on standard Euclidean space $\mathbb{E}^4$. Since these solutions are of finite-energy, they extend to the one point compactification of Euclidean space- S^4. The solutions with the lowest energy must, in fact, be either self dual or anti-self dual. To see this we start with the inequality

$$0 \leq \int_{\mathbb{E}^4} \mathrm{Tr}((F \pm \star F) \wedge \star (F \pm \star F)),$$

which leads to

$$S_{YM} = \int_{\mathbb{E}^4} \mathrm{Tr}(F \wedge \star F) \geq \left| \int_{\mathbb{E}^4} \mathrm{Tr}(F \wedge F) \right| = 8\pi^2 |c_2(E)|,$$

where $c_2(E)$ is the second Chern class of the bundle to which the connection belongs to. As the original inequality is saturated if and only if F is either self dual or anti-self dual, we have that the lowest energy solutions must correspond to exactly these. These solutions are called *instantons* and are of great mathematical and physical interest. We will restrict our attention to anti-self dual solutions as the theory for self dual solutions is analogous.

There are two twistoral methods for generating instanton solutions based on corollary 4.1; the method of algebraic curves and the ADHM construction. We will look at the former, specialising to the case of SU(2), as the twistor theory is more apparent than in the latter. It should be emphasised, however, that the ADHM construction is more general as it is readily adapted to many situations beyond SU(2) instantons. Furthermore, one can describe its method in an almost completely non-twistoral way, as in Manton & Sutcliffe [25], where much of the computation comes down to linear algebra.

Serre [26] showed that vector bundles over $\mathbb{CP}^3$ can naturally be associated with algebraic curves constructed from global sections of topologically twisted forms of said bundle. The construction of this association for our case is as follows.

Let E be a holomorphic rank-2 bundle satisfying the conditions in corollary 4.1. It can then be shown that $\check{H}^0(\mathbb{PT}, E) = 0$; that is, E possesses no global sections. So we instead look at the twisted bundle $E(l) = E \otimes L^l$, which will possess global sections for sufficiently large l. Suppose now that ψ is a global section of $E(l)$ and define

$$X = \{Z \in \mathbb{PT} |\ \psi(Z) = 0\}.$$

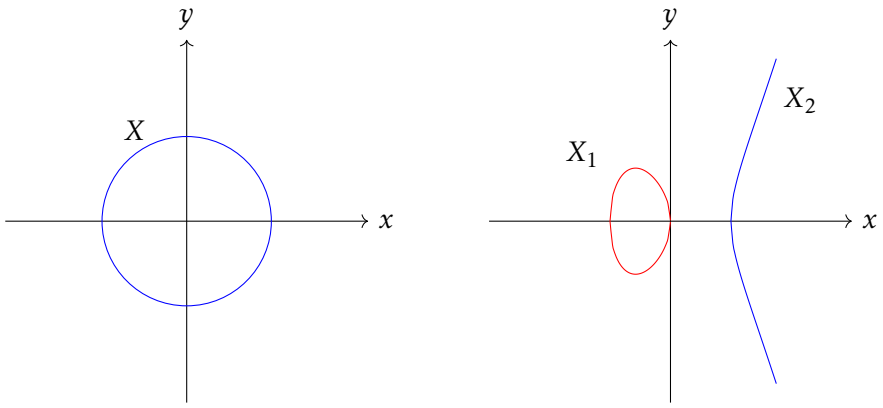


Figure 6: Two examples of algebraic curves, one connected and one disconnected. The curve X is given by $y^2 + x^2 = 4$, while the curve $X_1 \cup X_2$ is given by the relation $y^2 = x(x^2 - 4)$.

This set defines an algebraic curve in $\mathbb{PT}$, but X is not necessarily connected and could consist of r-many disconnected components X_i, as shown in figure 6. The degrees d_i and genera g_i of these components are, however, constrained by

$$\sum_{i=1}^{r} d_i = c_2(E) + l^2,$$
$$g_i - 1 = (n-2)d_i.$$

It is then clear that $\psi \in X$ is a nowhere vanishing global section of $E(l)$ over $\mathbb{PT} - X$, thus inducing a trivial line bundle I over $\mathbb{PT} - X$. Therefore, there exists the short exact sequence

$$0 \to I \to E(l) \to L^{2l} \to 0.$$

As a consequence, a transition matrix $F(Z^\alpha)$ of E can be written in the form

$$F(Z^\alpha) = \begin{pmatrix} \zeta^{-l} & \Gamma \\ 0 & \zeta^{l} \end{pmatrix}.$$

where $\zeta = \pi_{0'}/\pi_{1'}$ and $\Gamma \in \check{H}^1(\mathbb{PT} - X, \mathcal{O}(-2l))$. The observation of Atiyah and Ward [27] was that Γ defines a naturally anti-self dual solution ϕ to the Maxwell equations of helicity $(l-1)$ for $l \geq 2$ and a solution to the Laplace equation for $l = 1$, through the Penrose transform. Then by looking at the Penrose transform, we can form a sequence of ansätze $\{\mathcal{A}_l\}_{l=1}^{\infty}$ that take the zero-rest mass solution ϕ to an anti-self dual Yang-Mills field with the gauge group SU(2). Furthermore, Γ can be constructed exactly from the zero set X and r many non-zero elements of the curve data $\{d_i, g_i\}_{i=1}^{r}$, giving us a way to generate anti-self dual SU(2) connections from algebraic curves.

In order to construct the connection we need to find splitting matrices for $F(Z^\alpha)$ restricted to a line L_x. To do so, we expand $\Gamma(ix^{AA'}\pi_{A'}, \pi_{A'})$ as a Laurent series and define

$$\Gamma = \sum_{i \in \mathbb{Z}} \gamma_i \zeta^i = \sum_{i=-\infty}^{-1} \gamma_i \zeta^i + \gamma_0 + \sum_{i=1}^{\infty} \gamma_i \zeta^i := \Gamma_- + \gamma_0 + \Gamma_+,$$

where the coefficients γ_i are components of the zero rest mass field that Γ corresponds to. The splitting matrices then can be readily written down. Explicitly, the matrices

$$H = \frac{1}{\sqrt{\gamma_0}}\begin{pmatrix} \gamma_0 + \Gamma_- & \zeta^{-l}\Gamma_- \\ \zeta & 1 \end{pmatrix}, \quad \hat{H} = \frac{1}{\sqrt{\gamma_0}}\begin{pmatrix} -\zeta^{l}\Gamma_+ & -(\gamma_0 + \Gamma_+) \\ 1 & \zeta^{-l} \end{pmatrix}$$

satisfy $F = H\hat{H}^{-1}$ when restricted to a projective line L_x. We can now construct the connection from H and $\hat{H}$ as we did in section 4.3, from which we can simply state a sequence of ansätze $\{\mathcal{A}_l\}_{l=1}^{\infty}$ that generates an SU(2) instanton from the components of a ZRM field constructed from an algebraic curve.

Example 4.1. Let ϕ be a solution to Laplace's equation $\nabla^2\phi = 0$; that is, the scalar ZRM equation in Euclidean space. The ansätze $\mathcal{A}_1$ then gives an anti-self dual SU(2) connection of the form

$$A^a = i\bar{\sigma}^{ab}\nabla_b \log\phi,$$

where $\bar{\sigma}^{ab}$ is the anti-self dual bivector of Pauli matrices.

The ansätze $\mathcal{A}_1$ was known prior to the Atiyah-Ward construction but the higher helicity analogues were not. When generating $\mathcal{A}_2, \mathcal{A}_3, \ldots$, we notice that the ansätze become exceedingly complicated as the helicity increases, but the principle remains the same; zero-rest mass field in, SU(2) instanton out. Corrigan et al. [28] showed that $\mathcal{A}_{l+1}$ can be generated from $\mathcal{A}_l$ via a recurrence relation akin to a Bäcklund transformation, giving a systematic way of generating the full sequence and leading us to the main result.

Theorem 4.2. Let $l \geq 2$ and $\{\gamma_i\}_{i=1-l}^{l-1}$ be a set of holomorphic fields that satisfy

$$\nabla_{A0'}\gamma_i - \nabla_{A1'}\gamma_{i+1} = 0,$$

for $1-l \leq i \leq l-2$. Now let M be the $(k \times k)$ matrix with components $M_{rs} = \gamma_{r+s-l-1}$. Assume that $\det M \neq 0$ and define the following components of its inverse: $\alpha = (M^{-1})_{11}$, $\beta = (M^{-1})_{1k}$ and $\omega = (M^{-1})_{kk}$. Then the gauge potential defined component wise by

$$A_{B0'} = \frac{1}{2\alpha}\begin{pmatrix} \nabla_{B0'}\alpha & 0 \\ -2\nabla_{B1'}\omega & -\nabla_{B0'}\alpha \end{pmatrix},$$
$$A_{B1'} = \frac{1}{2\alpha}\begin{pmatrix} -\nabla_{B1'}\alpha & -2\nabla_{B0'}\beta \\ 0 & \nabla_{B1'}\alpha \end{pmatrix},$$

is an anti-self dual SU(2) connection.

Remark. It should be emphasised that the triviality and reality conditions that the bundle E should satisfy must be checked after the construction of E from a curve. This is due to the existence of *jumping lines* where $E|_{L_x}$ is not trivial, but it suffices to check that these don't occur on any real line.

4.5 Symmetry reductions

An interesting phenomenon occurs when we assume that the solutions to the ASDYM equations are invariant under the action of a subgroup of the complexified conformal group $\mathrm{GL}(4,\mathbb{C})$. To explore this, we follow two examples motivated by sections 5.1 and 6.3 of Mason & Woodhouse [29]. Let us assume that the anti-self dual gauge potential A has no x^4 dependence. Then we have the decomposition

$$A = \sum_{i=1}^{4} A_i \, \mathrm{d}x^i := -\phi \, \mathrm{d}x^4 + \boldsymbol{A},$$

where both ϕ and $\boldsymbol{A}$ are independent of t. The ASDYM equations in components can be written as

$$\epsilon_{abij} F^{ij} + 2\epsilon_{abi4} F^{i4} = -F_{ab},$$

where i, j run from 1 to 3 and a, b run from 1 to 4. The only important case occurs when a and b both lie within 1 and 3. Choosing a and b as such, leads to

$$2\epsilon_{ijk} \mathrm{D}^k \phi = F_{ij} \iff F = \star_3 \mathrm{D}\phi,$$

where $\star_3$ is the 3-dimensional Hodge star and we've noted that $F^{i4} = -\mathrm{D}^i\phi$. This is the ***Bogomolny equation***; it describes static magnetic monopoles and is a well known integrable system.

What we displayed is that a known integrable system can be regarded as a *reduction* of the ASDYM equations under some symmetry group. We can try this for any subgroup of $\mathrm{GL}(4,\mathbb{C})$ to get another integrable system. These reductions are then classified by:

1. A subgroup $H \subset \mathrm{GL}(4,\mathbb{C})$ for which the reduced solution is invariant under. In the Bogomolny reduction this is the group of translations in the x^4 direction.

2. The gauge group G of the system.

3. A real section used to reduce the complex system to a real one with possibly indefinite signature.

4. Jordan normal forms, or equivalently conjugacy classes, of *Higgs fields*. The Higgs fields ϕ_i of a reduction are defined by

   $$\phi_i = X_i \lrcorner A,$$

   where X_i are the generators of the group H. If H acts on $\mathbb{CM}^c$ freely, then a gauge transformation g satisfies $X(g) = 0$ and therefore

   $$\phi_i \mapsto g\phi_i g^{-1}.$$

   Through this we can place ϕ_i in Jordan normal form. If instead H does not act freely, then this property does not hold and we must adjust our classification.

Example 4.2. Now consider the reduction of the ASDYM equations by two translations X and Y. We can then subdivide the possible reductions by two translations depending on the behaviour of the metric on the surface $\Pi = \mathrm{span}\{X, Y\}$. There are four possible cases, but the two cases where the metric vanishes on Π are impossible in a Lorentzian spacetime so we choose to take X and Y such that the metric is reduced to rank 1 on Π.

By taking appropriate double-null coordinates $(w, \tilde{w}, z, \tilde{z})$, as in section 4.2, we can take X and Y to have the form

$$X = \partial_w - \partial_{\tilde{w}}, \quad Y = \partial_{\tilde{z}}.$$

Taking X and Y as such means that our gauge potential A is a function of $x = \frac{1}{2}(w + \tilde{w})$ and $t = z$ only. Additionally, by passing over to the quotient space $\mathbb{CM}^c/\Pi$ we can also make the identification

$$V(x,t) \sim V(x,t) + \alpha X + \beta Y.$$

for any vector $V(x,t)$. Using this can reduce the Lax pair for the ASDYM equations. Explicitly, we reduce L by noting

$$L = \mathrm{D}_{\tilde{z}} - \zeta \mathrm{D}_w \sim \mathrm{D}_{\tilde{z}} - \zeta \mathrm{D}_w - \tfrac{1}{2}\zeta X - Y,$$

which upon expansion gives us

$$L = A_{\tilde{z}} - \zeta(\partial_x + A_w).$$

A similar calculation tells us that M can be reduced to

$$M = \partial_x + A_{\tilde{w}} - \zeta(\partial_t + A_z).$$

We simplify the situation by using a gauge transformation to impose $A_{\tilde{w}} = 0$. The resulting reduced compatibility conditions in this gauge are then

$$\begin{gathered} \partial_x A_{\tilde{z}} = 0, \qquad \partial_x A_z - \partial_t A_w + [A_w, A_z] = 0, \\ \partial_x A_w + \partial_t A_{\tilde{z}} + [A_z, A_{\tilde{z}}] = 0. \end{gathered} \tag{4.3}$$

Now we take the gauge group to be $\mathrm{SL}(2,\mathbb{C})$. Therefore, A is an $\mathfrak{sl}(2,\mathbb{C})$ valued one-form, so A_z and such are (2×2) traceless matrices. There are two cases of importance.

The first case occurs when there exists a gauge such that

$$A_{\tilde{z}} = \begin{pmatrix} k & 0 \\ 0 & -k \end{pmatrix},$$

for some constant k. If such a gauge exists, then the reduced equations 4.3 tell us that A_w must have the form

$$A_w = \begin{pmatrix} 0 & \bar{\psi} \\ -\psi & 0 \end{pmatrix},$$

where ψ is some functions of x and t. Equations 4.3 also imply that

$$A_z = \begin{pmatrix} |\psi|^2 & \partial_x \bar{\psi} \\ \partial_x \psi & -|\psi|^2 \end{pmatrix},$$

with the conditions

$$-2k\partial_t \psi = \partial_x^2 \psi + 2\bar{\psi}|\psi|^2,$$
$$2k\partial_t \bar{\psi} = \partial_x^2 \bar{\psi} + 2\psi|\psi|^2.$$

Taking $k = i/2$ reduces this to a single equation; the *non-linear Schrödinger (NLS) equation.*

The second case occurs when $A_{\tilde{z}}$ is not diagonalisable. We can therefore find a gauge such that

$$A_{\tilde{z}} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

up to scaling. Following the same argument above, we arrive at the following form for A_z

$$A_z = \frac{1}{8}\begin{pmatrix} 4\partial_x(\partial_x v - v^2) & -8\partial_x v \\ \partial_x^3 v - 4v\partial_x^2 v - 2(\partial_x v)^2 + 4v^2\partial_x v & -4\partial_x(\partial_x v - v^2), \end{pmatrix}$$

where v is a function such that $u = -\partial_x v$ satisfies the *KdV equation*

$$4\partial_t u = \partial_x^3 u + 12u\partial_x^2 u.$$

Yet again we have reduced the ASDYM equations to two more systems which were well known to be integrable prior to the Penrose-Ward correspondence. One could continue to play this game of selecting subgroups of $\mathrm{GL}(4,\mathbb{C})$ and reducing the ASDYM equations, as in Ablowitz et al. [30], and each time arrive at a known integrable system. This leads us to *Ward's conjecture.*

Conjecture 4.1. (Ward [31]) Many (and perhaps all?) of the ordinary and partial differential equations that are regarded as being integrable or solvable may be obtained from the self-dual gauge field equations (or its generalisations) by reduction.

While it was shown that there are classes of integrable systems that do not arise as reductions of the ASDYM equations, the process of symmetry reduction gives us a unified appraoch to the integrability of the reduced systems through the Penrose-Ward correspondence. An area of interest is seeing how the Penrose-Ward correspondence relates to the lower dimensional integrable methods, such as the *inverse scattering transform* for the KdV and NLS equations, and how this can be adapted to other systems. We refer the interested reader to Mason & Woodhouse [29] for an analysis of this, as we do not have the time to go into full detail of both the inverse scattering transform and its relation to the Riemann-Hilbert problem found in the Penrose-Ward correspondence.

5 Twistors in curved spacetime

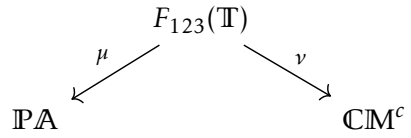
Finally, we will round out this report by detailing the adaption of twistor theory to curved spacetimes and the difficulties surrounding this. Historically, this was studied as early as 1968 in a paper by Penrose [32] but it was quickly seen that there has been no clear single way to generalise the results from flat spacetimes. However, for a special class of spacetimes there is a well studied way of generating a twistor correspondence which we aim to describe. This will then lead to a brief description of the twistor construction of gravitational instantons.

5.1 Approaches to generalisation

In pursuit of generalising twistor theory to curved backgrounds, we ask how exactly can we define the flat twistor space and how can this be adapted. We shall breifly detail some of these and discuss their viability.

The first possiblity is to regard twistors as spinors of the conformal group. This works well in flat space as we can compactify it to form a conformally symmetric domain, but this clearly doesn't adapt well to spacetimes with no conformal symmetry.

Perhaps the most complete definition for flat twistor space was through the double fibration picture we developed. However, there is a different double fibration that we could have developed that leads to a more adaptable correspondence. Our new double fibration takes the form:



where $\mathbb{PA} = F_{13}(\mathbb{T})$ is called the *projectified ambitwistor space*. Ambitwistor space is more a adaptable notion as it can be viewed as the space of complex null lines of $\mathbb{CM}^c$. Furthermore, we can in fact exhibit a double fibration picture which works for any manifold, which we will now construct.

Let $\mathcal{M}_0$ be some real manifold equipped with the metric g_0 and let $(\mathcal{M}, g)$ be the corresponding complexified spacetime. A *complex null direction* at a point $x \in \mathcal{M}$ is a one-form $\vartheta \in \mathrm{T}^*_x\mathcal{M}$ with vanishing norm $g^{-1}(\vartheta, \vartheta) = 0$. We therefore define the null subbundle of the cotangent bundle as

$$\mathrm{T}^*_N\mathcal{M} = \{(x, \vartheta) \in \mathrm{T}^*\mathcal{M} |\ g^{-1}(\vartheta, \vartheta) = 0\} \subset \mathrm{T}^*\mathcal{M}.$$

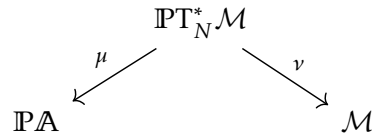
We now want to identify points that lie on the same null geodesic. To do so, we need to quotient the null subbundle by the action of the vector field

$$\partial_N = \vartheta^a \left(\frac{\partial}{\partial x^a} + \Gamma^c_{ab} \vartheta_c \frac{\partial}{\partial \vartheta_b} \right),$$

whose integral curves are lifts of geodesics to the cotangent bundle. These geodesics are naturally scaled however, so we will projectify the null subbundle by identifying $(x, \vartheta) \sim (x, \lambda\vartheta)$ for any non-zero $\lambda \in \mathbb{C}$ to arrive at

$$\mathbb{PA} = \mathbb{P}\mathrm{T}^*_N\mathbb{M}/\partial_N.$$

This also exhibits that there is a natural projection into ambitwistor space from the null subbundle. We therefore arrive at the double fibration



The major advantage of this approach it can be shown, as in section 2 of [33] for example, that $\mathbb{PA}$ has a natural complex structure in the sense that it is a 5-dimensional holomorphic symplectic manifold. Furthermore, as discussed in [33] again, gauge and gravitational fields on $\mathcal{M}$ can be interpreted as holomorphic data on ambitwistor space. Despite these attractive qualities, there has been little progress with this formulation in terms of fully realising curved twistor theory. Many of the results using ambitwistors are not as strong as those for flat twistor theory with the most major results actually pertaining to the flat case yet again.

The final generalisation we will discuss is the idea of projective twistor space being the space of α-planes in $\mathbb{CM}$. For curved spaces, we need an analogue to α-planes from which we can form the curved projective twistor space. The natural generalisation is an *α-surface* which we define to be a two-dimensional surface S with the property that all tangent vectors at each point are null and orthogonal. The existence of these α-surfaces depends on the underlying manifold and is governed by *Frobenius' theorem*, for which we now detail it and its preliminary definitions.

Definition 5.1. Let $\mathcal{M}$ be a manifold and E be a subbundle of $\mathrm{T}\mathcal{M}$. We say that E is *involutive* if for any E-valued vector fields U and V, the Lie bracket $[U, V]$ is also E-valued.

Definition 5.2. Let $\mathcal{M}$ be a manifold and E be a subbundle of $\mathrm{T}\mathcal{M}$. The subbundle E is said to be *integrable* if for each $x \in \mathcal{M}$ there exists a submanifold $\mathcal{N} \subset \mathcal{M}$ through x such that $\mathrm{T}\mathcal{N} = E|_{\mathcal{N}}$.

Theorem 5.1. (Frobenius) A subbundle E of the tangent bundle of some manifold $\mathcal{M}$ is involutive if and only if it is integrable.

Suppose now that we have a 4-dimensional manifold $\mathcal{M}$ that admits an α-surface through each point. This means that there exist independent null vectors v_x and w_x at each point which, additionally, are orthogonal. We now define the subbundle E of the tangent bundle by

$$E = \bigsqcup_{x \in \mathcal{M}} \mathrm{span}\{v_x, w_x\}.$$

This is simply a bundle of α-planes over the manifold. The subbundle E is integrable because the α-surfaces are exactly the submanifolds with tangent space equal to the restriction of E to each surface. Therefore, the Frobenius theorem tells us that E must be involutive and there must exist functions α and β such that

$$[v, w]^a = v^b \nabla_b w^a - w^b \nabla_b v^a = \alpha v^a + \beta w^a,$$

for v^a and w^a in E. As v^a and w^a are null and orthogonal they can be written in spinor form as $v^a = \lambda^A \pi^{A'}$ and $w^a = \xi^A \pi^{A'}$, which upon substution into the Frobenius condition gives

$$\pi^{A'} \nabla_{AA'} \pi_{B'} = \chi_A \pi_{B'}, \tag{5.1}$$

for some spinor χ_A. This imposes a constraint on the curvature of $\mathcal{M}$ so we shall briefly discuss the Riemann tensor in order to interpret this constraint.

The central quantity measuring the curvature of a spacetime $(\mathcal{M}, g)$ is the rank-$(0,4)$ Riemann tensor R_{abcd} defined, in a coordinate basis, by

$$R_{abc}{}^{d} w^c = \nabla_a \nabla_b w^d - \nabla_b \nabla_a w^d,$$

for any vector w^d. From the Riemann tensor we can form two lower rank tensors; the *Ricci tensor* $R_{ab} = R_{acb}{}^{c}$ and the *Ricci scalar* $R = R_a{}^a$. It is often helpful to decompose the Riemann tensor into sums of irreducible representations of the Lorentz group. In components this decomposition is given by

$$\begin{aligned} R_{abcd} = \Psi_{ABCD} \epsilon_{A'B'} \epsilon_{C'D'} + \tilde{\Psi}_{A'B'C'D'} \epsilon_{AB} \epsilon_{CD} \\ + \Phi_{ABC'D'} \epsilon_{A'B'} \epsilon_{CD} + \Phi_{A'B'CD} \epsilon_{AB} \epsilon_{C'D'} \\ + 2\Lambda (\epsilon_{AC} \epsilon_{BD} \epsilon_{A'B'} \epsilon_{C'D'} + \epsilon_{AB} \epsilon_{CD} \epsilon_{A'D'} \epsilon_{B'C'}), \end{aligned}$$

where $\Lambda = \frac{R}{24}$ is the *cosmological constant* of the spacetime. The first two terms in this decomposition are anti-self dual and self dual respectively and together form the *Weyl tensor* C_{abcd}

$$C_{abcd} = \Psi_{ABCD} \epsilon_{A'B'} \epsilon_{C'D'} + \tilde{\Psi}_{A'B'C'D'} \epsilon_{AB} \epsilon_{CD}.$$

The Weyl tensor represents the conformally invariant part of the Riemann tensor and as such only encodes information about the *conformal class* of the spacetime.

Now we see that applying $\pi^{B'} \pi^{C'} \nabla^{A}_{C'}$ to 5.1 introduces the self dual part of the Weyl tensor through the Ricci identity. Namely, we have that

$$\tilde{\Psi}_{A'B'C'D'} \pi^{A'} \pi^{B'} \pi^{C'} \pi^{D'} = 0, \tag{5.2}$$

is the integrability condition for the subbundle E. We aren't quite in analogy with flat twistor space yet, however. With this construction we have a family of α-surfaces parameterised by the spacetime point $x \in \mathcal{M}$ only, therefore giving us a 4 parameter family of surfaces. In the flat space case we had a 6 parameter family of surfaces, 2 for each complex coordinate of $\mathbb{CP}^3$, so we require two further parameters. This is obtained by stipulating that the surfaces are also parameterised by the spinor field $\pi_{A'}$, giving us an α-surface through each point for any $\pi_{A'}$. With this property, the integrability condition now becomes[11]

$$\tilde{\Psi}_{A'B'C'D'} = 0.$$

The Weyl tensor is therefore anti-self dual. Spacetimes with this property are said to be anti-self dual and admit a curved twistor correspondence which we will now explore.

[11] It is not necessary to redo the Frobenius argument to find the integrability condition as we can apply the same argument for each spinor field $\pi_{A'}$. The integrability condition is then 5.2 for each spinor field $\pi_{A'}$ as opposed to a specific spinor field.

5.2 The twistor theory of anti-self dual spacetimes

Let us now focus on an anti-self dual spacetime $(\mathcal{M}, g)$ and try to form a twistor correspondence between the set of such spaces and the moduli space of their α-surface structure. Our discussion is motivated by section 9.1 of Ward & Wells [10], where the full details of our construction can be found.

First of all, we require that the space of α-surfaces of an anti-self dual spacetime $\mathscr{P}$, which we will refer to as the twistor space of the spacetime, have a proper manifold structure. To do so, we can require the manifolds we consider to be *geodesically convex*. These space have the property that for any two points in a manifold there is a geodesic connecting them. This represents no local loss of information but it is actually easier to consider a subset of these manifolds called *civilised manifolds*. A manifold is said to be civilised if its twistor space is homeomorphic to $\boldsymbol{P^I}$, thus ensuring a nice manifold structure.

Now recall that in flat space, each point $x \in \mathbb{CM}$ corresponds to a projective line L_x in $\mathbb{PT}$. In curved spacetime we have an analogous correspondence, but this time for a different set of curves which we now define.

Definition 5.3. A *rational curve* X on a complex manifold $\mathcal{M}$ is a one-dimensional submanifold of $\mathcal{M}$ with the property that there exists a holomorphic isomorphism $f : X \to \mathbb{CP}$ whose inverse is also holomorphic. This function is called ***biholomorphic*** and X is said to be biholomorphic to $\mathbb{CP}$.

Any curved twistor space naturally has a four parameter family of rational curves, one for each point in $\mathcal{M}$, which we denote by $\{C_x\}$. To characterise these curves better it is instructive to look at the tangent bundles of the curves and the entire manifold itself through a local twistor prescription.

Let Z and Y be points in $\mathscr{P}$ connected by an infinitesimal vector εQ. In the spacetime picture these correspond to α-surfaces S_Z and S_Y connected by a vector field εv^a. The connecting vector field is not unique in light of this correspondence, however, as we can add any vector tangent to S_Z and regain the same points in twistor space. We therefore define the non-tangential part of v^a as $\alpha^A = iv^{AA'}\pi_{A'}$ where $\pi_{A'}$ is the spinor field associated to S_Z.

Since εv^a connects S_Z and S_Y, it follows that the change of v^a along S_Z is tangent to S_Z. Therefore, if $\pi^{A'}$ is the spinor field associated to S_Z and χ^A is some spinor field on the surface we have

$$\mathcal{L}_{S_Z} v^a = \chi^B \pi^{B'} \nabla_{BB'} v^{AA'} - v^{BB'} \nabla_{BB'} (\chi^A \pi^{A'}) = \xi^A \pi^{A'},$$

for some spinor $\xi^{A'}$ and $\mathcal{L}_{S_Z}$ denotes the Lie derivative along S_Z. Hence

$$\pi_{A'} \chi^B \pi^{B'} \nabla_{BB'} v^{AA'} = \chi^A \beta^{A'} \pi_{A'},$$

where $\beta^{A'} = v^b \nabla_b \pi^{A'}$. The crucial point is that the pair $(\alpha^A, \beta_{A'})$, called a ***local twistor*** in this context, defines the vector Q in twistor space. Furthermore, by manipulating the equations involving α^A and $\beta^{A'}$ we obtain the following propogation equations

$$\chi^B \pi^{B'} \nabla_{BB'} \alpha^A = -i \chi^A \pi^{B'} \beta_{B'},$$
$$\chi^B \pi^{B'} \nabla_{BB'} \beta_{A'} = -\chi^B \pi^{B'} \alpha^A P_{ABA'B'},$$

where $P_{ab} = -\frac{1}{2}R_{ab} + 2\Lambda g_{ab}$ is the *Schouten tensor*. The equations determine the pair of spinor fields $(\alpha^A, \beta_{A'})$ over S_Z given some initial data and as such are called the *equations of local twistor transport*. However, the local twistor description has a redundancy which we must factor out. Namely, the zero vector $Q = 0$ is represented by non-zero local twistors of the form $\kappa(0, \pi_{A'})$ for some constant κ so all local twistors should be understood modulo these zero terms.

Since a rational curve C_p is biholomorphic to $\mathbb{CP}$ we may use $\pi_{A'}$ as homogeneous coordinates parameterising the curve and any vector field on it. As such, let $Q = Q(\pi_{B'})$ be a vector field over C_p which corresponds to a local twistor field. We evaluate the local twistor field at the point p to obtain the local twistor $(\Omega^A(\pi_{B'}), \Pi_{A'}(\pi_{B'}))$ which corresponds to Q tangent to the curve C_p. It is clear that both Ω^A and $\Pi_{A'}$ are homogenous of degree one in $\pi_{B'}$, so we have that $\mathrm{T}(\mathscr{P})|_{C_p}$ consists of a pair of spinors homogenous of degree 1, up to the zero terms discussed above. On the other hand, vectors tangent to C_p itself correspond to local twistors of the form $(0, \Pi_{A'})$, up to zero terms. We therefore have

$$\mathrm{N}(C_p) = \mathrm{T}(\mathscr{P})|_{C_p}/\mathrm{T}(C_p) \cong \{\Omega^A(\pi_{B'})|\ \Omega^A(k\pi_{B'}) = k\Omega^A(\pi_{B'})\} \cong L^1 \oplus L^1.$$

This is how we characterise these curves; each rational curve in $\mathscr{P}$ that corresponds to a spacetime point $x \in \mathcal{M}$ has *normal bundle* $\mathrm{N}(C_p) \cong L^1 \oplus L^1$. This leads us to the central theorem of this section; the *non-linear graviton construction*.

Theorem 5.2. (Penrose [34]) There is a one-to-one correspondence between:

1. civilised anti-self dual spacetimes $(\mathcal{M}, g)$, up to conformal scaling;
2. three dimensional complex manifolds $\mathscr{P}$ which contain a four parameter family of rational curves, each with normal bundle $L^1 \oplus L^1$, and are homeomorphic to the flat finite projective twistor space $\boldsymbol{P^I}$.

We have already shown one direction but we will not prove going from 2 to 1. For a full proof see the original paper [34] or Ward & Wells [10].

The non-linear graviton construction only determines an anti-self dual spacetime up to conformal scaling if we are given a curved twistor space. We can, in fact, exactly determine the metric of the spacetime in such a way that the *Einstein condition* $R_{ab} = \frac{1}{4}Rg_{ab}$ holds. This is done, as we will see, by endowing the twistor space with some additional information that then constrains the spacetime further.

Let us first see how we can impose the Einstein condition. We've shown that any vector Q at the point $Z \in \mathscr{P}$ corresponds to a local twistor $(\alpha^A, \beta_{A'})$. This allows us to define the quantity

$$\tau(Q) = \pi_{B'}\beta^{B'},$$

where $\pi_{B'}$ is the spinor field associated to S_Z. Importantly, it can be easily seen that $\tau(Q) \neq 0$ for any Q tangent to a rational curve C_p. However, for τ to be well defined, we need it to be constant on S_Z, which gives us the condition

$$\chi^B\pi^{B'}\nabla_{BB'}(\pi^{A'}\beta^{A'}) = i\chi^A\alpha^A\pi^{A'}\pi^{B'}\Phi_{ABA'B'} = 0,$$

where $\Phi_{ab} = R_{ab} - \frac{1}{4}Rg_{ab}$. We therefore conclude that the existence of τ imposes that the spacetime $(\mathcal{M}, g)$ is Einstein. But what exactly is τ? Well, as τ maps vectors to a number it is a differential form. There is a caveat however; it is homogeneous of degree 2 in $\pi_{A'}$ whereas true differential forms are homogeneous of degree 0. This means that we should view τ as a *twisted form*; that is, a section of the bundle $\Omega^1(\mathscr{P})\otimes\mathscr{L}^2$ where the bundle $\mathscr{L}^k$ is the analogue of the bundle L^k for the surface S_Z.

The other information on $\mathscr{P}$ required is not as obvious as τ, so let us state the corollary and then motivate the other requirements from the opposite direction.

Corollary 5.1. (Ward [35]) There is a one-to-one correspondence between:

1. civilised anti-self dual Einstein spacetimes $(\mathcal{M}, g)$ with cosmological constant Λ;
2. three dimensional complex manifolds $\mathscr{P}$, homeomorphic to $\boldsymbol{P^I}$, such that
   - there exists a four parameter family of rational curves C_x, each with normal bundle $L^1 \oplus L^1$;
   - there exists a holomorphic line bundle $\mathfrak{L}$ over $\mathscr{P}$ such that $\mathfrak{L}|_{C_x} \cong L^{-1}$;
   - there is a holomorphic one-form τ homogeneous of degree two, such that $\tau(Q) \neq 0$, for any non-zero Q tangent to any curve C_x;
   - there is a holomorphic three-form ρ homogeneous of degree four, such that $\tau \wedge \mathrm{d}\tau = 2\Lambda\rho$.

Remark. It should be noted that the requirement that there should be a line bundle $\mathfrak{L}$ such that $\mathfrak{L}|_{C_p} = L^{-1}$ is somewhat unneeded. This is because we can always construct it as the fourth root of the canonical line bundle $K = \bigwedge^3 \mathrm{T}^*\mathscr{P}$ so we will not dwell on this point.

Say that we have a twistor space $\mathscr{P}$ satisfying the conditions of corollary 5.1. The basic idea that we used to construct the metric on $\mathcal{M}$ is that our twisted forms each define a symplectic form on the spin bundles over $\mathcal{M}$ which in turn determines the metric. In order to do this, however, we need to first identify what our spin spaces are. To this end, we define

$$V = \{Q \in \mathrm{T}\mathscr{P} |\ \tau(Q) = 0\} \subset \mathrm{T}\mathscr{P},$$

which is easily seen to be of rank-2 and satisfies $V|_{C_p} \cong \mathrm{N}(C_p)$. Observing that $V|_{C_p}\otimes L^{-1} \cong I\oplus I$, we define the spin spaces to be

$$S_p^- = \Gamma(C_p, V|_{C_p} \otimes L^{-1}),$$
$$S_p^+ = \Gamma(C_p, L^1).$$

It is clear that $\mathrm{T}_p\mathcal{M} \cong S_p^-\otimes S_p^+$, so these definitions are consistent with the notion of spin spaces.

Now let ξ and μ be elements of the primed spin space S_p^+. Note that they are, by definition, homogeneous of degree 1 so we conclude that $\xi\,\mathrm{d}\mu - \mu\,\mathrm{d}\xi$ is homogeneous of degree 2. Since C_p is 1-dimensional, we must have

$$\xi\,\mathrm{d}\mu - \mu\,\mathrm{d}\xi = \varepsilon_+(\xi, \mu)\tau|_{C_p},$$

for some two-form ε_+. Since $\tau(Q) \neq 0$ for Q tangent to C_p, we have that ε_+ is well defined and constitutes a symplectic form on the primed spin space.

The construction for S_p^- is similar. Let ξ and μ be elements of the unprimed spin space S_p^-. The symplectic form ε_- is then defined by

$$\varepsilon_-(\xi, \mu)\tau(\nu) = 2\rho(\nu, \xi, \mu),$$

for some vector field ν on C_p such that $\tau(\nu) \neq 0$. The symplectic form ε_- is particularly simple, being homogeneous of degree 0 and independent of τ meaning it is in fact constant.

With the two symplectic forms defined properly, the metric is simple to write down. Namely, the metric g is given by $g = \varepsilon^- \otimes \varepsilon^+$. The only thing left is to show that the curvature is indeed 24Λ, but this is long and ultimately comes down to defining an appropriate connection and checking its properities. We therefore refer the reader to [35] for full details of the proof.

We will not explicity construct any spacetimes through corollary 5.1, as examples are readily available in Ward & Wells [10] and the construction is reasonably evident from our discussion above, but we shall mention a particularly interesting case.

As in the case of the ASDYM eqauations, we can use the extension of the non-linear graviton construction to generate a graviational analogue to instantons. These solutions, called *graviational instantons*, are either compact and without a boundary or look asymptotically flat (in the case of Riemannian anti-self dual spacetimes, they obviously must look asymptotically Euclidean). Of course, anti-self dual solutions are but a subset of graviational instantons, but many examples of them may be generated this way. Indeed, Hitchin [36] showed that the twistoral construction can recreate the Gibbons-Hawking multi-instanton while Ward [35] showed that $\mathbb{CP}^2$ with the Fubini-Study metric can be constructed as an instanton twistorally as well.

6 Conclusion

With the end of our discussion of curved twistors, we have came to the close of this report. We started by looking at the underlying mathematics of twistor theory, forming a theory which could simply be viewed as mathematically striking before showing the applications of the theory. Indeed, the ZRM equations and ASDYM equations showed us the undeniable utility of twistors in the theory of integrability, even leading to the quite surprising unification of certain integrable methods through symmetry reductions. We then showed that twistors could give a concise and geometric way of constructing instantons, verifying the physical usefulness of the theory. Finally, the curved twistor space discussion gave us a glimpse towards what Penrose had originally envisioned for twistor theory; a theory of quantum gravity. While we did not delve into any quantum mechanics or quantum field theory, we were able to start forming a twistor theory for certain curved spaces in the hope that its quantised theory leads to promising results.

In this report we have seen several situations where twistor theory succeeds in an unprecedented manner and leads to groundbreaking results like the Penrose-Ward correspondence. On the other hand, we've also started to see some of its shortcomings like its inability to easily generalise to general spacetimes or general gauge fields. It is also worth noting that while we were able to start generalising our theory to curved spaces, we are still none the wiser to how quantum gravity can emerge from twistor theory. These shortcomings, however, do not lessen the importance of the theory as research in to twistor theory has continued with much success.

One of the most important modern developments in twistor theory came in 2003 when Witten [37] showed that several results from the study of scattering in supersymmetric gauge theories could be reinterpreted as a string theory with the target space an extension of twistor space. Witten's paper led to a flurry of research into the so-called twistor string theory that is still an extremely active area of research today. Furthermore, the ideas of Witten have been applied to other theories to give new results or replicate old ones in a simple and intuitive manner. Skinner [38], for example, gave a worldsheet interpretation of $\mathcal{N} = 8$ supergravity and through a collaboration with Cachazo [39] gave new results for scattering in this theory.

Of course, research into the more traditional aspects of twistor theory has not ceased. For example, Hitchin et al. [40] established that a twistor correspondence exists for any hyperkähler manifold and used this to construct hyperkähler metrics. Furthermore, Penrose [41] recently presented a new revision of twistor theory, called palatial twistor theory, with the hope of resolving some of the long standing problems of twistor theory, such as the googly problem.

All in all, we've seen the humble beginnings of twistor theory start to expand into unexpected, yet fascinating, areas of both mathematics and physics. Given the surprising nature of twistor theory, it is hard to think where its next major application may be, but with advances in both standard twistor theory and twistor string theory it is clear that it will be of great value for many years to come.

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